Portfolio Substitution and the Effects of Bank Capital Regulation

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Abstract

A large literature finds that bank capital requirements reduce lending, but these studies generally estimate lending responses to regulations that change required equity for multiple bank assets simultaneously. For example, a 1-percentage-point increase in a bank's risk-based capital requirement raises the required equity on risk-weight-1 loans by 1 percentage point and on risk-weight-0.2 securities by 0.2 percentage points. Evaluating counterfactual policies that impose a new pattern of equity requirements across assets, such as changing a single asset's risk weight, requires estimating the response of an asset's supply to its own regulatory cost while holding other assets' costs fixed. I develop a portfolio-allocation model and empirical method to estimate each asset's own-cost elasticity of supply using changes in bank-level capital requirements. The model uses the fact that portfolio shares sum to one to express each asset's own-cost response and portfolio substitution in terms of the same elasticities. I recover positive supply elasticities for every asset category, with securities holdings responding more to regulatory costs than loans do. Applying these elasticities to post-crisis requirements, I ask whether stronger capital regulation explains large banks' shift from lending to securities since the financial crisis. Capital regulation accounts for about 28 percent of the aggregate shift toward securities and 35 percent of the cross-sectional convergence in large banks' average asset risk weights since 2011.

1 Introduction

Bank capital requirements are regulators' principal tool for protecting financial stability. Capital requirements require banks to finance a minimum share of their assets with equity, reducing banks' incentives to take risks. But stronger capital requirements also reduce lending by raising banks' funding costs (e.g., Aiyar et al., 2014; Bridges et al., 2014; Gropp et al., 2019; Berrospide and Edge, 2019; Juelsrud and Wold, 2020; Favara et al., 2021; Blattner et al., 2023).

Banks' funding costs can rise when banks finance a larger share of their assets with equity, in contrast to the Modigliani–Miller benchmark in which a firm's financing mix leaves its total funding cost unchanged, because deposits and other safe bank debt provide liquidity services that investors value, allowing banks to borrow at lower rates (Gorton and Pennacchi, 1990; Stein, 2012; DeAngelo and Stulz, 2015; Gorton and Winton, 2017). Understanding how capital requirements affect banks' portfolio composition is therefore central to assessing the costs and benefits of capital regulation.

Since the financial crisis, regulators have strengthened capital requirements for large U.S. banks, while these banks have simultaneously shifted their portfolios away from lending and toward securities (Hanson et al., 2024). A prominent explanation for banks' declining role in credit provision is that stronger bank regulation has discouraged bank lending and encouraged lending by nonbanks (Buchak et al., 2018; Irani et al., 2021; Lee et al., 2024).

Because past changes in bank capital requirements have typically altered the equity requirements on multiple assets at once, most studies estimate how an asset's supply responds when its own equity requirement and other assets' requirements change together. For example, raising a bank's required risk-based capital (RBC) ratio by 1 percentage point, from 10 to 11 percent, raises the bank's equity requirement for risk-weight-1 loans by 1 percentage point, for risk-weight-0.2 securities by 0.2 percentage points, and so on. In contrast, the supplementary leverage ratio (SLR) requires 5 percent equity financing for all assets, requiring more equity funding against securities and less against loans than the RBC rule. Research shows that introducing the SLR led affected banks to reduce their holdings and intermediation of securities (Duffie and Krishnamurthy, 2016; Favara et al., 2022; Bräuning and Stein, 2024). But since the SLR simultaneously raises required equity on securities and lowers it on loans compared to the RBC, this estimated response reflects both channels. Separating them requires variation in the cost of one asset class holding the other fixed, which is the focus of this paper.

More broadly, predicting banks' response to a counterfactual policy that imposes different capital requirements across assets than those observed under current policies is a general

problem for regulators. For example, suppose that regulators are concerned about risks in commercial real estate and consider raising equity requirements for commercial real estate alone, leaving other assets' equity requirements unchanged. Since this policy has not occurred in the data, predicting its effects would require using historical changes in regulation that have shifted the equity requirements on multiple assets at once (as is usually the case) and empirically separating an asset's direct response to its own regulatory cost from the bank's portfolio substitution in response to changes in other assets' costs.

To solve this problem, I develop a portfolio-allocation model and empirical method to estimate each asset's own-cost elasticity of supply using past changes in bank-level capital requirements that altered the equity requirements of multiple assets at once. In the model, each asset's own-cost elasticity governs both its direct response to its own regulatory cost and its substitution response to changes in other assets' regulatory costs. The portfolio adding-up identity links these responses across assets, so each asset's substitution response depends on the other assets' elasticities.

My method recovers positive supply elasticity estimates for every asset category, meaning that each asset's portfolio share falls in response to an increase in its own equity requirement, holding other assets' requirements fixed. Securities respond more to their own regulatory costs than loans do. This is intuitive because banks can purchase securities in liquid markets, while expanding lending requires finding and screening borrowers. The model's predictions match out-of-sample bank responses to changes in capital requirements.

The resulting supply elasticity estimates are structural parameters useful for counterfactual policy analysis. I use them to ask whether stronger capital regulation explains large banks' shift from lending to securities since the financial crisis. Applying the estimated asset supply elasticities to post-crisis capital requirements, I find that capital regulation accounts for about 28 percent of the aggregate shift from lending toward securities and 35 percent of the cross-sectional convergence in large banks' average asset risk weights since 2011. Stronger capital requirements thus explain part, but not most, of the decline in large banks' lending shares and the convergence in their average asset risk weights.

This paper begins by developing a model of bank portfolio allocation across multiple types of loans and securities. For each asset, the model imposes the same local curvature in net-return functions across banks while allowing net-return levels to differ, so banks share a common local own-cost elasticity of supply for a given asset but can earn different returns on the same asset. In this model, holding return schedules and bank sizes fixed, an asset loses portfolio share when its regulatory cost rises relative to the average regulatory cost of all the bank's assets, weighted by their supply elasticities. The intuition follows from the portfolio adding-up identity: since portfolio shares sum to one, an expansion in one asset's

portfolio share requires a contraction in other assets' shares. The same own-cost elasticity governs both an asset's direct response to its regulatory cost and its substitution response to changes in other assets' regulatory costs.

A feature of the model is that it is straightforward to estimate in the data. I estimate asset supply elasticities using changes in bank-level capital requirements that alter assets' regulatory penalties by different amounts. The sample covers 2011–2025 and includes 28 domestic bank holding companies ("banks") with more than $100 billion in assets in at least one quarter during this period. The analysis covers banking-book assets only, excluding the trading book. I match quarterly bank balance-sheet data from Form FR Y-9C to the standardized asset risk weights prescribed by U.S. capital regulation (12 CFR Part 217, Subpart D), grouping assets into four risk-weight categories: 0, 0.2, 0.5, and 1. I combine these risk weights with bank-level capital requirements, which vary by bank and over time, to construct a measure of bank–asset–quarter equity requirements. I estimate a supply elasticity for each asset category by relating changes in banks' portfolio shares in each asset to regulatory changes to the required equity for each asset, pooling observations across banks and regulatory-change quarters.

My method identifies asset supply elasticities using two sources of regulatory variation: increases in risk-based capital requirements and switches in banks' binding capital rule. The risk-based capital (RBC) rule requires more equity against assets with higher regulatory risk weights, while the supplementary leverage ratio (SLR) requires the same equity share for all assets, regardless of asset risk. Large banks must satisfy both rules; a bank's binding rule is the rule that requires the most equity funding for its portfolio. When the RBC binds, an increase in the bank's required RBC ratio raises its required equity more for assets with higher risk weights. Since 2013, the Fed has raised RBC requirements by different amounts for different banks over time. Switches between binding rules can instead raise a bank's required equity on some assets while lowering it on others. For example, the introduction of the SLR caused custodian and dealer banks to switch from being RBC-bound to SLR-bound, raising their effective equity requirement on securities and lowering it on loans; subsequent increases in RBC requirements caused several of these banks to switch back.

I use both sources of variation to compare 4–5-year portfolio-share growth across assets facing different changes in required equity within the same bank, relative to portfolio-share growth in the same assets at other banks. I estimate all asset supply elasticities jointly by nonlinear least squares, controlling for asset–date–horizon fixed effects to absorb shocks to assets' net returns that are common to all banks holding the same asset. Identification relies on a parallel-trends assumption, as in difference-in-differences. The assumption is that, absent changes in regulation, differences in the portfolio-share growth across assets at the

same bank would have been the same as those at other banks.

My results show that correcting for portfolio substitution is necessary to recover the magnitude and sign of each asset's own-cost elasticity of supply. For every risk-weight category, I find that increasing an asset's equity requirement, holding other assets' equity requirements fixed, reduces that same asset's portfolio share. Correcting for portfolio substitution makes the estimated portfolio-share response more negative in every asset category. Without the substitution correction, the estimated response for risk-weight-0.5 assets, mostly first-lien mortgages, has the wrong sign: mortgages appear to gain portfolio share when their own equity requirement rises. Banks can shift toward mortgages even when mortgage equity requirements rise if securities' equity requirements rise by even more.

The estimated asset supply elasticities imply that securities respond more to their own regulatory costs than loans do. This is consistent with securities' greater liquidity and homogeneity: a bank can purchase securities in a standardized, liquid market, whereas expanding lending requires finding and screening additional borrowers and may also require building borrower relationships and developing expertise in unfamiliar markets. These factors limit how easily a bank can increase lending when its regulatory cost falls.

To illustrate the magnitude of the implied portfolio responses, consider imposing a 5 percent SLR equity requirement on a bank previously bound by a 10 percent RBC equity requirement, so that the SLR becomes binding. The SLR imposes a flat equity requirement across all assets regardless of risk, so it raises required equity on low-risk-weight securities and lowers it on high-risk-weight loans. Applying the estimated supply elasticities to these asset-specific changes in required equity implies a decline of about 3.2 percentage points in the portfolio share of risk-weight-0 assets (mostly Treasury securities and some reverse repo) and an increase of about 1.7 percentage points in the portfolio share of unsecured loans (e.g., credit card loans and C&I).

I evaluate the model's portfolio-share predictions out of sample using the 2020 and 2021 changes in Stress Capital Buffers, which add bank-specific equity requirements based on Federal Reserve stress-test results. These changes were excluded from the elasticity estimation. The model-implied portfolio-share changes match the direction of the observed changes in three of the four asset categories.

Beyond estimating asset-level own-cost elasticities of supply, I use the model to explain banks' persistent specialization in different assets through differences in the net returns that banks earn on the same asset. I use the estimated asset supply elasticities to translate cross-bank differences in portfolio shares into implied differences in net returns, after accounting for equity requirements. The estimates reveal that traditional banks have a comparative advantage in lending, while dealers and custodians have a comparative advantage in securities

and reverse repos. These comparative advantages are worth several percentage points in annual net returns.

Finally, I examine the contribution of stronger capital requirements to large banks' shift from lending to securities since 2011, an issue central to the debate over banks' declining role in credit provision (Hanson et al., 2024; Buchak et al., 2018; Irani et al., 2021; Lee et al., 2024; Buchak et al., 2024). Using the actual equity requirements that large banks faced on every asset risk-weight category in each quarter since 2011, I compute banks' model-implied portfolio shares. This partial-equilibrium exercise holds banks' sizes and assets' net-return schedules fixed. I find that stronger capital requirements account for about 28 percent of large U.S. banks' observed shift from lending to securities between 2011 and 2024.

I also estimate the contribution of post-crisis capital requirements to the convergence in large banks' average asset risk weights since 2011. Since each bank tends to be bound by the rule that most strongly penalizes its core assets, risk-based capital requirements encourage lending specialists to expand securities holdings, while the leverage requirement encourages dealers and custodians to expand lending (Greenwood et al., 2017). I find that post-crisis capital requirements account for about 35 percent of the observed convergence in banks' average portfolio risk weights.

The remainder of the paper proceeds as follows. Section 2 develops the model of banks' choice of portfolio allocation and equity ratio subject to regulatory capital requirements and derives the estimating equation. Section 3 describes the data, regulatory variation, and empirical implementation. Section 4 presents the own-cost supply elasticity estimates, examines alternative specifications, validates the model's predictions using out-of-sample changes in capital requirements, and recovers banks' comparative advantages. Section 5 estimates the contribution of post-crisis capital requirements to large banks' shift from lending toward securities and to the convergence in their portfolio compositions. Section 6 concludes.

Contribution to the literature. Most studies of bank capital requirements estimate how lending responds when equity requirements on loans and other assets change together. My paper estimates how an asset's supply responds to its own regulatory cost when other assets' regulatory costs are held fixed.

The model's portfolio-substitution mechanism also helps interpret existing evidence that balance-sheet constraints limit banks' market intermediation. This literature links balance-sheet constraints such as leverage regulation to reduced repo and Treasury-market intermediation and persistent deviations from covered interest parity (Duffie and Krishnamurthy, 2016; He et al., 2022; Duffie et al., 2023; Bräuning and Stein, 2024; Du et al., 2018). My estimates show that the SLR shifts banks' banking-book portfolios away from securities and toward lending. The model illustrates how substitution toward lending can reinforce the

direct effect of higher equity requirements on securities holdings.

My paper also contributes to the literature on banks' portfolio allocation and substitution across assets. Furfine (2001) estimates a dynamic model of bank portfolio choice under capital requirements and regulatory monitoring. Wang et al. (2022) structurally estimate banks' joint lending, securities, and funding choices under capital and reserve requirements to study monetary policy transmission. Li et al. (2026) model banks' joint choices of securities holdings, corporate lending, lending to nonbanks, and deposit funding, subject to a balance-sheet constraint. My estimation framework is most closely related to Diamond et al. (2024), who use instruments for loan demand, deposit demand, and reserve supply to estimate how banks' marginal costs depend on their balance-sheet composition. My approach instead uses variation in regulatory costs and the portfolio adding-up identity to separately estimate own-cost supply elasticities and substitution across securities and loan categories.

My framework studies how differences in asset risk weights shape allocation across a bank's portfolio as a whole, complementing studies of how loan-specific capital requirements shape allocation within corporate or mortgage lending (Behn et al., 2016; Mayordomo and Rodríguez-Moreno, 2018; Dietsch et al., 2020; Plosser and Santos, 2024). Within corporate lending, Fraisse et al. (2020) use differences in capital requirements across bank–firm pairs to show that higher loan-specific requirements reduce lending. Within mortgage lending, Benetton et al. (2021) use variation in capital requirements across loans in the same lender's mortgage book to show that lower requirements reduce mortgage rates; Benetton (2021) estimates the effects of capital requirements on mortgage origination costs and lender competition.

This paper also contributes to the debate over banks' declining role in credit provision. Begenau and Landvoigt (2022) show that higher bank capital requirements expand shadow banking, while Acharya et al. (2026) argue that bank regulation encourages banks to replace direct lending to borrowers with financing for nonbanks that lend to those borrowers. Sundaresan and Xiao (2024) identify a related effect of liquidity regulation on banks' portfolios: liquidity regulations encourage banks to hold cash and liquid securities at the expense of lending. Chernenko et al. (2025) challenge the capital-requirements explanation for private credit's growth and emphasize bank supervision and the regulatory treatment of indirect lending. Technology and funding offer additional explanations: Buchak et al. (2024) find that improvements in securitization made securities-based financing more attractive to borrowers relative to bank balance-sheet loans, while savers' growing preference for securities over deposits reduced bank funding. Hanson et al. (2024) emphasize the role of deposit growth in banks' declining loan shares. Koont and Walz (2026) find that banks' lending response to deposit inflows has weakened as balance-sheet expansion has become more costly.

My exercise isolates the contribution of capital requirements to portfolio composition, holding bank size fixed, and finds that stronger capital requirements explain about 28 percent of large banks' shift from lending toward securities.

Finally, my estimation builds on demand systems that link own-price and cross-price responses through budget constraints and structural assumptions (Brainard and Tobin, 1968; Deaton and Muellbauer, 1980). Koijen and Yogo (2019) develop an instrumental-variables approach to estimating asset demand. Fuchs et al. (2023) show that misspecified substitution can distort estimated own-price elasticities. Haddad et al. (2026) show that, under conditional homogeneity of substitution, cross-sectional designs identify relative elasticities but leave substitution patterns needed for broader counterfactuals unidentified. I apply this approach to banks' asset supply, using regulatory changes that move several assets' funding costs simultaneously. I assume that each asset's net-return function depends only on its own portfolio share. Together with the portfolio adding-up identity, this assumption links an asset's direct response to required equity to its substitution responses to other assets' costs, allowing me to estimate supply elasticities jointly from portfolio-share responses.

2 Model and Estimation

The model provides an estimating equation for asset-level own-cost elasticities of supply. I first describe the bank's problem and optimality conditions, then use the portfolio adding-up identity to derive how an asset's portfolio share responds to changes in its own equity requirement and those of other assets.

2.1 Bank Portfolio Choice

Banks choose their portfolio composition and equity ratios subject to risk-based capital (RBC) requirements and a supplementary leverage ratio (SLR). Under RBC, assets with higher legally prescribed risk weights w_a require more equity financing, and banks with higher required ratios of equity to risk-weighted assets ℓ_{bt}^{RBC} must finance more of each asset with equity. The SLR requires the same equity financing share for every asset, with ℓ_{bt}^{SLR} denoting a bank's required ratio of equity to total assets.[1]

Bank b chooses portfolio shares s_{abt} in assets a and an equity-to-assets ratio E_{bt} in period t, holding total assets fixed. The index a covers all assets in the bank's portfolio.[2] The function

[1]The statutory SLR measures Tier 1 capital as a share of total leverage exposure, which includes off-balance-sheet exposures. Here its required ratio is converted to a requirement per dollar of assets, holding leverage exposure relative to assets fixed.

[2]In the empirical implementation, assets without an observed standardized risk weight are grouped into

$r_{abt}(s_{abt})$ gives the bank's expected income from asset a, net of operating and debt funding costs, divided by the bank's total assets. Net returns are separable across assets. I assume that r_{abt} is continuously differentiable and strictly concave in the asset's portfolio share. Diminishing marginal net returns can reflect declining marginal revenues, rising marginal operating costs, or both. For example, expanding lending may require paying higher wages to recruit additional loan officers or offering lower loan rates to attract additional borrowers.

Banks' funding costs can rise when banks finance a larger share of their assets with equity, in contrast to the Modigliani–Miller benchmark in which a firm's financing mix leaves its total funding cost unchanged, because banks' deposits and other safe, short-term debt provide liquidity services that investors value, allowing banks to issue these claims at lower financing costs (Gorton and Pennacchi, 1990; Stein, 2012; DeAngelo and Stulz, 2015; Gorton and Winton, 2017). The parameter c_E measures the additional cost per dollar of financing assets with equity rather than debt.

The bank maximizes expected profit subject to portfolio and capital constraints:

$$\begin{aligned} &\max_{\{s_{abt}\},E_{bt}} \quad \sum_a r_{abt}(s_{abt}) - c_E E_{bt} \\ &\text{subject to} \quad \sum_a s_{abt} = 1, \quad s_{abt} \geq 0, \quad E_{bt} \geq \ell_{bt}^{RBC} \sum_a w_a s_{abt}, \quad E_{bt} \geq \ell_{bt}^{SLR} \sum_a s_{abt}. \end{aligned} \tag{1}$$

I derive the portfolio first-order conditions for an interior allocation, with $s_{abt} > 0$ for every asset. The Lagrangian incorporates the portfolio adding-up identity with multiplier η_{bt} and the capital constraints with multipliers $\lambda_{bt}^{RBC}, \lambda_{bt}^{SLR} \geq 0$:

$$\begin{aligned} \mathcal{L}_{bt} = &\sum_a r_{abt}(s_{abt}) - c_E E_{bt} + \eta_{bt} \left(1 - \sum_a s_{abt}\right) \\ &+ \lambda_{bt}^{RBC} \left(E_{bt} - \ell_{bt}^{RBC} \sum_a w_a s_{abt}\right) + \lambda_{bt}^{SLR} \left(E_{bt} - \ell_{bt}^{SLR} \sum_a s_{abt}\right). \end{aligned}$$

The first-order condition for equity is

$$\lambda_{bt}^{RBC} + \lambda_{bt}^{SLR} = c_E.$$

When equity is more costly than debt, the bank chooses the minimum equity ratio that satisfies both capital requirements. Ruling out the knife-edge case in which RBC and SLR require exactly the same amount of equity financing for the bank's portfolio, the rule that

an outside good and retained in the portfolio-share denominator. They are included in the asset index here. The baseline estimation holds their combined portfolio share fixed, so changes in the shares of the remaining asset categories sum to zero; Appendix C relaxes this restriction.

requires more equity binds and the other rule is slack. By complementary slackness, the slack requirement's multiplier is zero, so the binding requirement's multiplier equals c_E.

Bank b's marginal equity requirement on asset a in quarter t, k_{abt}^{Bind}, is the partial derivative of its required equity ratio with respect to asset a's portfolio share, holding the other asset shares fixed. In the sums below, a' indexes all assets in the bank's portfolio:

$$\begin{aligned} k_{abt}^{Bind} &= \frac{\partial}{\partial s_{abt}} \max \left\{ \ell_{bt}^{RBC} \sum_{a'} w_{a'} s_{a'bt},\ \ell_{bt}^{SLR} \sum_{a'} s_{a'bt} \right\} \\ &= \mathbf{1}\{\text{RBC binds}\} \ell_{bt}^{RBC} w_a + \mathbf{1}\{\text{SLR binds}\} \ell_{bt}^{SLR}. \end{aligned} \tag{2}$$

The equity requirement raises the funding cost of asset a by $c_E k_{abt}^{Bind}$ per dollar compared with financing the asset entirely with debt.

The marginal equity requirement can differ across banks for the same asset because different capital rules bind. For example, suppose Capital One holds only credit-card loans and Goldman Sachs holds only Treasuries.[3] Both banks face a 10 percent RBC requirement and a 5 percent SLR requirement. Capital One's portfolio of credit-card loans with a risk weight of 1 requires 10 percent equity financing under RBC and 5 percent equity financing under SLR. Goldman Sachs's Treasury portfolio requires 0 percent equity financing under RBC but 5 percent equity financing under SLR. Each bank must satisfy the larger of its two equity requirements, so Capital One is constrained by the RBC and maintains equity equal to 10 percent of risk-weighted assets, while Goldman Sachs is constrained by the SLR and maintains equity equal to 5 percent of total assets. Capital One therefore faces marginal equity requirements of 10 percent on credit-card loans and 0 percent on Treasuries, while Goldman Sachs faces 5 percent on both assets.

The asset first-order condition equates marginal net returns to equity funding costs and the opportunity cost of displacing other assets:

$$r'_{abt}(s_{abt}) = c_E k_{abt}^{Bind} + \eta_{bt}. \tag{3}$$

Expanding one asset's portfolio share requires reducing other assets' shares. The multiplier η_{bt} captures the opportunity cost of this displacement, in addition to the direct equity funding cost $c_E k_{abt}^{Bind}$. If the equity funding premium is zero ($c_E = 0$), capital requirements do not affect the bank's asset allocation. If equity is costly, an increase in an asset's equity requirement raises its marginal funding cost. The bank responds by adjusting its portfolio allocation.

[3]This example is adapted from Greenwood et al. (2017).

2.2 Portfolio Responses to Equity Requirements

An asset's portfolio-share response reflects both its own regulatory cost and substitution induced by changes in other assets' costs. I derive a locally linear relationship between changes in equity requirements and portfolio shares to estimate local own-cost elasticities of asset supply. I assume that r_{abt} is twice continuously differentiable near a fixed portfolio share $\bar{s}_{ab}$, with $r''_{abt}(\bar{s}_{ab}) = -1/\tau_a < 0$. The curvature $1/\tau_a$ is common across banks for each asset and fixed over time. Its inverse, τ_a, measures the asset's local own-cost elasticity of supply. Let Δ denote a change from period $t-1$ to period t, and let $\Delta r'_{abt}(\bar{s}_{ab})$ denote the change in marginal net returns evaluated at the same reference share. Locally approximating marginal net returns and taking differences of the first-order condition gives

$$\Delta s_{abt} \approx \tau_a \left(\Delta r'_{abt}(\bar{s}_{ab}) - c_E \Delta k^{Bind}_{abt} - \Delta \eta_{bt} \right). \tag{4}$$

Appendix A.1 derives equation (4) using a second-order Taylor expansion of net returns around the reference portfolio share $\bar{s}_{ab}$.

The local own-cost supply elasticity τ_a determines the direct response of asset a's portfolio share to a change in its regulatory funding cost $c_E k^{Bind}_{abt}$, holding net-return schedules, other assets' regulatory costs, and the portfolio opportunity cost η_{bt} fixed. The portfolio opportunity cost adjusts when equity requirements change, because the bank must reallocate the displaced portfolio share to other assets. The Lagrange multiplier η_{bt} ensures that portfolio shares sum to one. Holding net-return schedules and bank size fixed, an identical change in the binding equity requirement, k^{Bind}_{abt}, for every asset leaves portfolio shares unchanged.

Proposition 1 shows how the portfolio adding-up identity links each asset's response to its own regulatory cost to its substitution response to changes in other assets' costs.

Proposition 1 (Portfolio substitution). *Suppose net-return schedules and bank size are fixed. Consider small portfolio adjustments around fixed asset shares $\bar{s}_{ab}$. Assume the bank continues to hold a positive amount of every modeled asset and faces one binding capital constraint and one slack capital constraint. The change in an asset's portfolio share in response to changes in required equity across assets is*

$$\Delta s_{abt} \approx -c_E \tau_a \left(\Delta k^{Bind}_{abt} - \Delta \bar{k}^{\tau}_{bt} \right), \qquad \Delta \bar{k}^{\tau}_{bt} = \frac{\sum_a \tau_a \Delta k^{Bind}_{abt}}{\sum_a \tau_a}. \tag{5}$$

Holding net-return schedules and bank size fixed, an asset loses portfolio share when its equity requirement rises by more than the supply-elasticity-weighted average increase in equity requirements across the bank's assets. Since portfolio shares sum to one, an expansion in one asset's portfolio share requires a contraction in other assets' shares. In equation (5),

the own-cost component is $-c_E\tau_a\Delta k^{Bind}_{abt}$, and the substitution component is $c_E\tau_a\Delta\bar{k}^{\tau}_{bt}$. Each asset's substitution response depends on the supply elasticities of all assets in the portfolio. The adding-up identity therefore links the own-cost and substitution responses across assets. I use these links to estimate the supply elasticities jointly while accounting for portfolio substitution. Appendix A proves Proposition 1.

2.3 Estimating Own-Cost Supply Elasticities

I estimate each asset's own-cost elasticity of supply, τ_a, from portfolio-share responses to regulatory changes in equity requirements, k^{Bind}_{abt}, calibrating the equity funding premium to $c_E = 0.05$ using estimates in the literature (Appendix C.1). This calibration translates changes in required equity into changes in funding costs.

To allow portfolio shares to adjust over time, I estimate local projections of portfolio-share changes from quarter $t-1$ to quarter $t+h$. Following Proposition 1, the explanatory variable is the change in the asset's equity requirement from quarter $t-1$ to quarter t, net of the supply-elasticity-weighted average change in equity requirements across the bank's assets:

$$s_{ab,t+h} - s_{ab,t-1} = \delta^{(h)}_{at} + \beta_a \left(\Delta k^{Bind}_{abt} - \Delta\bar{k}^{\tau}_{bt}\right) + u^{(h)}_{abt}. \tag{6}$$

Here $\Delta k^{Bind}_{abt} = k^{Bind}_{abt} - k^{Bind}_{ab,t-1}$, and h is the number of quarters after the change in the asset's equity requirement.

The estimating equation includes asset–date–horizon fixed effects, $\delta^{(h)}_{at}$, to absorb portfolio-share responses to asset–date shocks in marginal net returns: the model's common supply elasticity for each asset makes these responses common across banks. Bank–date shocks that shift marginal net returns equally across all of a bank's assets cancel through the portfolio adding-up identity, leaving portfolio shares unchanged. I assume that the remaining bank–asset–date return shock, $\Delta\varepsilon_{abt}$, is uncorrelated with regulatory changes in equity requirements across the bank's assets, conditional on the fixed effects. The residual $u^{(h)}_{abt}$ captures portfolio-share responses to bank–asset return shocks and the approximation error from linearizing marginal net returns. Appendix A.2 derives estimating equation (6) from the bank's first-order conditions.

Comparing the coefficient on $\Delta k^{Bind}_{abt} - \Delta\bar{k}^{\tau}_{bt}$ in the model's portfolio-share response, equation (5), with its coefficient in the estimating equation (6) gives $\beta_a = -c_E\tau_a$. Given the calibrated equity funding premium, the estimated response coefficient therefore recovers the own-cost elasticity of supply:

$$\tau_a = -\frac{\beta_a}{c_E}.$$

Asset a's substitution response, $c_E \tau_a \Delta \bar{k}^{\tau}_{bt}$, depends on its own supply elasticity τ_a and on all assets' supply elasticities through the weights in $\Delta \bar{k}^{\tau}_{bt}$. I therefore estimate all assets' supply elasticities jointly by nonlinear least squares. Under the model's common equity funding premium c_E, the weighted-average funding-cost change $c_E \Delta \bar{k}^{\tau}_{bt}$ is common across a bank's assets. The corresponding asset-level substitution response $c_E \tau_a \Delta \bar{k}^{\tau}_{bt}$ varies with the asset's supply elasticity τ_a, so bank–date fixed effects do not generally absorb portfolio substitution. For each candidate set of elasticities, I compute the weighted-average change in required equity, $\Delta \bar{k}^{\tau}_{bt}$, and the predicted portfolio share change, $-c_E \tau_a (\Delta k^{Bind}_{abt} - \Delta \bar{k}^{\tau}_{bt})$, for each asset. Predicted portfolio share changes sum to zero within each bank.

3 Data and Regulatory Variation

3.1 Data and Sample

Data and sample. I use quarterly FR Y-9C data on bank balance sheets and capital ratios for 28 domestic bank holding companies that exceed $100 billion in assets in at least one quarter during the sample.[4] The estimation sample covers 2011–2025.[5] I combine bank-level capital requirements with asset risk weights to measure the equity required per dollar of each asset at each bank and match these requirements to quarterly portfolio shares. I restrict the analysis to assets in a bank's banking book and exclude trading assets from the portfolio-share denominator. I also exclude reserves and vault cash from the assets used to calculate portfolio shares to abstract from surges or contractions in bank reserves driven by quantitative easing or quantitative tightening.

Risk-weight mapping. I match banks' banking-book assets reported on the FR Y-9C to four standardized risk-weight categories—0, 0.2, 0.5, and 1—using the weights prescribed by the U.S. capital rule implementing Basel III (12 CFR Part 217, Subpart D). Table 1 summarizes the asset mapping. I use the labels RW0, RW0.2, RW0.5, and RW1 for these categories, respectively. Appendix Table B.2 lists the FR Y-9C items, assigned risk weights, and legal provisions underlying the mapping. I calculate portfolio shares relative to total banking-book assets, excluding reserves and vault cash. Assets that cannot be assigned to these four risk-weight categories remain in the portfolio-share denominator. The baseline

[4] I treat banks before and after large mergers (those in which the target exceeds 25 percent of the acquirer's assets at close) as separate entities so that within-bank changes do not span the acquisitions. Appendix Table B.1 lists the acquisitions. In the estimation below, I cluster standard errors by bank rather than merger-split entity, giving 28 clusters.

[5] The dynamic response estimates use starting quarters from 2011:Q2 through 2024:Q4; the long-run estimates use starting quarters from 2011:Q2 through 2021:Q4. Portfolio outcomes are observed through 2025:Q4, with pre-period outcomes available from 2011:Q1.

estimation holds their combined portfolio share fixed. Appendix C allows substitution into these remaining assets and estimates their supply elasticity alongside those of the four categories. The estimates for the four categories remain similar to the baseline estimates on the same samples.[6]

Binding rule. A bank's binding rule is the rule that requires the bank to hold the most Tier 1 capital given its portfolio composition and regulatory exposures. Equivalently, it is the rule against which the bank has the smallest Tier 1 capital buffer, measured as actual minus required Tier 1 capital.[7] Before banks began reporting the SLR on the FR Y-9C in 2016:Q3, I use a proxy constructed from FR Y-9C data; Appendix B describes its construction.

I smooth short-lived switches in the binding-rule classification before constructing each asset's required equity. When a bank briefly switches away from a rule and then returns to it, I assign the surrounding rule to the intervening quarters. The procedure generally removes reversals lasting up to four quarters, but preserves spells lasting at least three quarters that begin during the temporary pandemic exemption of reserves and Treasuries from the SLR denominator.[8] As a separate robustness check, Appendix C.13 allows banks to be uncertain about which rule binds and to place continuous weight on both RBC and SLR requirements. I jointly estimate the supply elasticities τ_a and a parameter governing how these weights depend on the difference between the bank's capital buffers under the SLR and RBC requirements. Each buffer is actual Tier 1 capital minus the amount required under the corresponding rule, divided by risk-weighted assets.

3.2 Regulatory Changes and Identifying Variation

I identify asset supply elasticities from two sources of regulatory variation: increases in risk-based capital requirements and switches in banks' binding capital rule. Asset risk weights determine how each change in a bank-level capital requirement translates into changes in the equity required for individual assets. An increase in a binding risk-based capital requirement raises required equity more for assets with higher risk weights. A change in which rule binds can instead raise required equity on some assets while lowering it on others.

Changes to risk-based capital requirements. Changes in the minimum risk-based

[6]See Appendix B.2 for loan coverage and Appendix B.3 for reporting corrections.

[7]The Tier 1 leverage requirement occasionally binds instead of the RBC or SLR. I treat it as a separate binding requirement in all estimation. Observations bound by the Tier 1 leverage requirement are excluded from the illustrative figures. Banks also face separate risk-based requirements for common equity Tier 1 (CET1) and total capital; see 12 CFR 217.10. The binding-rule classification used here compares Tier 1 capital requirements and does not separately identify whether the CET1 or total-capital requirement is binding.

[8]Appendix B.3 gives the full smoothing procedure.

Tier 1 capital ratio, required Tier 1 capital buffers above that minimum, and bank-specific surcharges generate variation in banks' risk-based capital requirements over the sample period, as shown in Figure 1. Basel III raised the minimum Tier 1 capital requirement from 4 percent to 6 percent and introduced a Capital Conservation Buffer for all banks. Additional requirements for the eight largest and most complex banks (G-SIBs) vary over time with their size and systemic importance. The Capital Conservation Buffer and G-SIB surcharges were phased in from 2016 through 2019. Since 2020, banks subject to the Federal Reserve's stress tests must meet a Stress Capital Buffer that varies with each bank's projected losses under stress.[9] I exclude every bank-quarter with a change in the Stress Capital Buffer from the starting events used to estimate asset supply responses. Portfolio-share observations from those quarters remain available as outcomes for other starting events. Section 4.3 uses the 2020 and 2021 Stress Capital Buffer changes to evaluate the estimated supply elasticities.

Changes in banks' binding rule. The introduction of the SLR caused custodian and dealer banks to switch from being constrained by risk-based capital (RBC) requirements to being constrained by the SLR. Subsequent increases in risk-based requirements and the temporary pandemic SLR exemption caused some banks to switch back. Figure 2 shows that custodians such as State Street and dealers such as Goldman Sachs are usually SLR-bound, while lenders such as Capital One are usually RBC-bound. Custodians and dealers hold large amounts of assets with low risk weights, for which the SLR generally requires more equity than RBC. Lenders hold more high-risk-weight loans, making RBC the stronger requirement. During the pandemic, the temporary exclusion of reserves and Treasuries from the SLR reduced the equity required under the SLR, making even BNY Mellon and State Street RBC-constrained. Goldman Sachs also became RBC-constrained during the later part of the sample as its RBC requirement rose substantially, as shown in Figure 1. A sufficiently large increase in the RBC ratio can make RBC the stricter rule even for a bank whose portfolio is usually SLR-bound.

Asset-level required equity. I combine each bank's binding capital requirement with asset risk weights to construct the equity required per dollar for each bank–asset pair in each quarter, k_{abt}^{Bind}, as defined in equation (2). I estimate portfolio-share responses to changes in required equity per dollar of an asset. Both increases in risk-based capital requirements and switches in a bank's binding rule generate variation in this measure. For an RBC-bound bank, required equity equals the bank's required RBC ratio multiplied by the asset's risk

[9] I date the increase in the minimum ratio to the Basel III announcement in 2013:Q3. The annual Capital Conservation Buffer increment was 0.625 percentage points, reaching 2.5 percent. In 2020:Q4, the Stress Capital Buffer replaced that fixed buffer for covered banks, subject to the same 2.5 percent floor. See the Federal Reserve's 2013 Basel III rule, buffer transition provisions, 2015 G-SIB surcharge rule, and 2020 SCB rule.

weight. For an SLR-bound bank, the equity requirement is uniform across assets, except for assets covered by the temporary pandemic exemption. Table 2 reports these asset-specific requirements by bank in 2024:Q4. RBC-bound Capital One faced requirements of 11.5 percent for credit-card loans, 2.3 percent for agency mortgage-backed securities, and 0 percent for Treasuries, while SLR-bound Goldman Sachs faced 5 percent for every asset.

Figure 3 plots these asset-specific equity requirements over time for a lender, Capital One, and a dealer, Goldman Sachs.

Both sources of regulatory variation change equity requirements on multiple assets at once. Figure 4 illustrates how an increase in the RBC ratio and a switch in a bank's binding rule both change equity requirements across assets. On the left, for an RBC-bound bank, a one-percentage-point increase in the RBC ratio raises required equity by 1 percentage point for RW1 assets, 0.5 for RW0.5 assets, and 0.2 for RW0.2 assets, while leaving the equity requirement on RW0 assets unchanged. On the right, when a 5 percent SLR replaces a 10 percent RBC requirement as the binding constraint, required equity rises on low-risk-weight assets and falls on RW1 assets. In both cases, each asset's portfolio share reflects its direct response to its own equity requirement and portfolio substitution in response to changes in other assets' equity requirements.

I compare portfolio-share growth across assets facing different changes in required equity within the same bank, and compare those growth differences across banks holding the same assets. Identification relies on a parallel-trends assumption, as in difference-in-differences. The assumption is that, absent changes in regulation, differences in the portfolio-share growth across assets at the same bank would have been the same as those at other banks.

4 Estimation Results

I estimate own-cost supply elasticities using equation (6) and the regulatory variation in Section 3. For each risk-weight category, I find that raising an asset's equity requirement reduces its portfolio share when other assets' equity requirements are held fixed. Correcting for portfolio substitution makes the estimated portfolio-share response to an asset's own equity requirement more negative in every risk-weight category. For RW0.5 assets (mostly first-lien mortgages), the uncorrected estimate would instead imply that their portfolio share rises when their equity requirement increases. The corrected estimate implies that their portfolio share falls.

The estimated asset supply elasticities imply that securities respond more to their own regulatory costs than loans do. The common own-cost supply elasticity for RW0 and RW0.2 assets (such as Treasuries, agency securities, and interbank assets) is significantly larger than

the elasticity for RW1 assets (such as C&I and credit-card loans) at the 5 percent level. The difference between the common RW0/RW0.2 elasticity and the elasticity for RW0.5 assets (mostly first-lien mortgages) is statistically insignificant.

4.1 Own-Cost Supply Elasticity Estimates

Long-run portfolio-share responses. Figure 5 reports estimates of an asset's portfolio-share response to a one-percentage-point increase in its own equity requirement, pooling observations from quarters 16–20 after the increase, with and without the substitution correction. Correcting for portfolio substitution makes the estimated portfolio-share response more negative in every risk-weight category. For RW0.5 assets, the estimated portfolio-share response to a higher own-equity requirement is positive in the uncorrected regression and negative after substitution correction. The model implies that banks shift toward RW0.5 assets when their equity requirement rises by less than the elasticity-weighted average increase across all four asset categories. In the data, when required equity on RW0.5 assets rises, required equity on RW0 and RW0.2 assets tends to rise by even more. Banks then shed low-risk-weight securities and shift toward RW0.5 loans even though the equity requirement on RW0.5 loans has also risen. The uncorrected regression wrongly attributes the increase in the RW0.5 portfolio share to the higher equity requirement on RW0.5 loans.[10]

Portfolio-share responses over time. Figure 6 traces how an asset's portfolio share responds over time to a one-percentage-point increase in its equity requirement in a base quarter, holding other assets' equity requirements fixed. I estimate equation (6) separately at different quarterly horizons. For most asset categories, the substitution-corrected portfolio-share responses plateau over quarters 16–20 (Appendix A.3).

The baseline specification assumes that RW0 assets (such as Treasuries) and RW0.2 assets (such as agency securities and interbank assets) share a common own-cost supply elasticity, $\tau_0 = \tau_{0.2}$, and estimates separate elasticities for RW0.5 mortgages and RW1 loans. I make this assumption because, outside the temporary exemption of reserves and Treasuries from the SLR during the pandemic, changes in equity requirements across assets are always proportional to asset risk weights, and the portfolio-share changes for RW0, RW0.2, RW0.5,

[10]Both regressions include asset–date–horizon fixed effects. The uncorrected regression also includes bank–date fixed effects, so it compares portfolio-share changes across assets within a bank and date. Consider a bank switching from a binding 10 percent RBC requirement to a binding 5 percent SLR requirement. Required equity rises on low-risk-weight securities and falls on RW1 loans. In equation (5), the elasticity-weighted average change in required equity is about 2.9 percentage points. Because securities are more elastic, the substitution term offsets more of the securities-share decline than it adds to the RW1 share increase. The uncorrected regression reads the narrower gap in portfolio-share growth as a weaker response to each asset's own equity requirement. In this sample, binding-rule switches account for most of the modeled upward bias in the coefficients on assets' own equity requirements in each risk-weight category.

and RW1 sum to zero, leaving only three independent asset responses to estimate. The regulatory variation therefore identifies at most three free elasticity parameters, and imposing a common elasticity for RW0 and RW0.2 reduces the number of estimated elasticities from four to three.

From portfolio-share responses to own-cost supply elasticities. The model's portfolio-share response in equation (5) and the main regression in equation (6) imply $\widehat{\tau}_a = -\widehat{\beta}_a/c_E$. The equity funding premium calibrated in Section 2.3, $c_E = 0.05$, converts a one-percentage-point increase in required equity into a 5-basis-point increase in annual funding costs. Figure 7 and Table 3 report the own-cost supply elasticities implied by regressions of portfolio shares on equity requirements reported in Figure 5, with and without the substitution correction.

The own-cost supply elasticity for RW0 and RW0.2 assets is larger than the elasticities for RW0.5 and RW1 assets, consistent with securities' greater liquidity and homogeneity. Banks can purchase securities in liquid markets, while expanding lending requires finding and screening additional borrowers. The difference in own-cost supply elasticities between the pooled RW0 and RW0.2 categories and RW1 is statistically significant at the 5 percent level ($p = 0.031$); the difference between the pooled RW0 and RW0.2 categories and RW0.5 (mostly first-lien mortgages) is statistically insignificant ($p = 0.118$) (Appendix Table C.2).

Magnitude of the implied portfolio-share responses. To illustrate the magnitudes of the estimated elasticities, consider a bank switching from a binding 10 percent RBC requirement to a binding 5 percent SLR requirement, holding net-return schedules and total assets fixed. Under RBC, required equity equals the 10 percent bank-level requirement multiplied by the asset's risk weight, while under SLR, required equity is 5 percent for every asset. Compared with the 10 percent RBC requirement, the 5 percent SLR raises the equity requirement on low-risk-weight securities and interbank lending (RW0 and RW0.2) and lowers it on high-risk-weight loans (RW1). The equity requirement on first-lien mortgages (RW0.5) remains the same under either rule. The weighted-average increase in required equity across all four asset categories (RW0, RW0.2, RW0.5, and RW1), using the substitution-corrected supply elasticities in Table 3 as weights, is:

$$\Delta\bar{k}^{\tau} = \frac{(30.5 \times 5.0) + (30.5 \times 3.0) + (11.6 \times 0.0) + (4.4 \times (-5.0))}{30.5 + 30.5 + 11.6 + 4.4} \approx 2.9.$$

Switching from a binding 10 percent RBC requirement to a binding 5 percent SLR requirement increases the bank's required equity on government-backed securities (RW0) by $5.0 - 2.9 = 2.1$ percentage points net of the elasticity-weighted average increase across all assets. Given the calibrated equity funding premium $c_E = 0.05$ (Appendix Table C.1) and

the estimated RW0 elasticity of 30.5, the model implies a 3.2-percentage-point reduction in the RW0 portfolio share.

$$\Delta s_0 \approx -\frac{5}{100} \times 30.5 \times (5.0 - 2.9) \approx -3.2 \text{ percentage points.}$$

For RW1 loans (such as C&I and credit-card loans), switching from a binding 10 percent RBC requirement to a binding 5 percent SLR requirement reduces the bank's required equity by 7.9 percentage points net of the elasticity-weighted average increase across all assets. Given the same equity funding premium and the estimated RW1 elasticity of 4.4, the model implies a 1.7-percentage-point increase in the RW1 portfolio share.

$$\Delta s_1 \approx -\frac{5}{100} \times 4.4 \times (-5.0 - 2.9) \approx +1.7 \text{ percentage points.}$$

The bank's switch in binding rule leaves the equity requirement on RW0.5 mortgages unchanged at 5 percent, but the requirement falls by 2.9 percentage points relative to the elasticity-weighted average across assets. Given the estimated elasticity of 11.6 for RW0.5, the model implies that the mortgage portfolio share rises by about 1.7 percentage points. The higher equity requirements on RW0 and RW0.2 assets, which have the largest estimated supply elasticities, lead the bank to reduce their portfolio shares and reallocate toward other assets, including RW0.5 mortgages. The mortgage portfolio share thus rises purely through portfolio substitution.

$$\Delta s_{0.5} \approx -\frac{5}{100} \times 11.6 \times (0.0 - 2.9) \approx +1.7 \text{ percentage points.}$$

The RW0.5 coefficient in the uncorrected regression (Figure 5) is positive because banks shift toward mortgages when equity requirements rise more for low-risk-weight securities. Switches between binding RBC and SLR requirements account for most of the upward bias in that coefficient.

4.2 Robustness Checks

Identification relies on a parallel-trends assumption, as in difference-in-differences. The assumption is that, absent changes in regulation, differences in the portfolio-share growth across assets at the same bank would have been the same as those at other banks.

One potential concern is that a bank's choice of portfolio composition could potentially change which capital rule binds, which in turn determines the bank's marginal equity requirement on each asset. For example, an increase in demand for credit-card loans at one

bank could raise the bank's credit-card portfolio share and make RBC bind, raising the bank's marginal equity requirement on credit-card loans. Asset–date–horizon fixed effects absorb credit-card demand shifts that are common across banks, but a demand increase specific to one bank can still raise both its credit-card portfolio share and its required equity, biasing the estimated credit-card supply elasticity toward zero. However, if the bank was experiencing a rise in credit-card demand that caused it to expand credit-card lending for some time, the pretrend plots would show a rise in the bank's credit-card portfolio share before its required equity changed, and I find no statistically significant pretrends for any asset category (Figure C.1). I also check the sensitivity of my elasticity estimates to earlier changes in portfolio composition by controlling for lagged portfolio shares (Table C.3).

To rule out changes in required equity driven by the bank's chosen portfolio composition, I reestimate the baseline regressions using only changes in required equity that arise from changes in the bank's regulatory requirements, holding portfolio composition fixed. To do this, I take the bank's portfolio from the quarter before the regulatory event and calculate required equity for each asset twice. The first calculation uses the bank's RBC and SLR requirements from the preceding quarter. The second uses the bank's RBC and SLR requirements from the event quarter. I determine which rule binds for the fixed portfolio in each calculation and use the difference in required equity as the regressor. The resulting own-cost supply elasticity estimates are smaller than the baseline estimates for every asset category. In both sets of estimates, the common elasticity for RW0 and RW0.2 is largest, followed by RW0.5 and then RW1 (Table C.4).

Another potential concern is that my elasticity estimates might be sensitive to the inclusion of the temporary exemption of reserves and Treasuries from the SLR during the pandemic, because banks may respond differently to temporary and permanent changes in capital requirements. The pandemic SLR exemption over 2020–2021 also coincided with unusual economic conditions. To assess sensitivity to the exemption, I reestimate the supply elasticities excluding the introduction and expiration of the exemption from the set of regulatory events. The supply elasticity estimates are larger and less precise than in the baseline, and their signs are unchanged. This is consistent with banks responding less strongly to temporary changes in capital requirements than to permanent ones (Figure C.2 and Table C.5).

The estimated long-run response of an asset's portfolio share to its equity requirement might also capture the effects of regulatory changes that occurred before or after the base quarter. I therefore reestimate the baseline regressions, controlling for required-equity changes before and after the base quarter to separate their effects from those of the base-quarter regulatory change (Table C.6). This yields point estimates similar to the baseline elasticities, so I use the baseline elasticities in the policy application exercise of Section 5.

Another potential concern is that the statistical significance of the estimated supply elasticities may be sensitive to correlation in regression errors across banks and to the small number of bank clusters. The baseline regression's standard errors are clustered by bank, with 28 banks in the sample. I report standard errors clustered by both bank and quarter to allow for cross-bank correlation, and because conventional clustered tests may be inaccurate with few clusters, I also report wild cluster bootstrap inference alongside the conventional tests (Table C.7).

Another potential concern is that my finding that supply elasticities are larger for securities than for loans could be sensitive to assumptions about which asset categories have a common elasticity. The baseline estimates separate elasticities for RW0.5 and RW1 loans while pooling RW0 and RW0.2 assets. Appendix Table C.8 reports the supply elasticities and standard errors from the baseline specification, a two-group specification that estimates one elasticity for RW0 and RW0.2 assets and a separate elasticity for RW0.5 and RW1 assets, and a specification in which elasticities vary linearly with risk weight. The estimated supply elasticities remain positive and larger for securities than for loans.

Banks may also adjust their holdings of banking-book assets outside the four modeled categories. The baseline specification holds those assets' portfolio share fixed. I relax this restriction by estimating a supply elasticity for those assets as a fifth category in Table C.9. The elasticity estimates for the four modeled categories are similar to the baseline estimates reestimated on the same observations.

To test the sensitivity of my supply elasticity estimates to possible misclassification of asset risk weights in the mapping in Appendix Table B.2, I compare those estimates with estimates based on banks' reported allocations of assets to each risk-weight category in Schedule HC-R, the regulatory capital schedule of the FR Y-9C. The detailed breakdown is available from 2015 onward, so I restrict the comparison to regulatory changes from 2015 onward and use the same bank observations for both specifications (Table C.10). The resulting supply elasticity estimates are larger than but qualitatively similar to the baseline estimates on the same sample.

I also assess how sensitive the own-cost supply elasticity estimates are to the assumed equity funding premium. If the true annual equity funding premium exceeds the baseline value of 5 percent, each percentage-point increase in required equity raises funding costs by more than the baseline calibration assumes. A given estimated portfolio-share response would then imply a smaller own-cost supply elasticity. An equity funding premium below 5 percent would imply a larger own-cost supply elasticity (Figure C.4).

To assess sensitivity to the assumption that only one capital rule binds for a bank in a given quarter, Appendix C.13 allows banks to place continuous weight on both RBC and

SLR requirements. I jointly estimate the parameter governing the regulatory weights and the supply elasticities. The resulting own-cost supply elasticities remain positive and larger for securities than for loans, but are less precise than the baseline estimates.

4.3 Out-of-Sample Validation

To test whether the baseline supply elasticities generalize to other regulatory changes, I compare model-implied and observed portfolio-share changes following the 2020 and 2021 Stress Capital Buffer (SCB) changes, which I excluded from the regulatory events used to estimate the baseline elasticities.[11] The model matches the direction of the observed portfolio-share changes in response to the SCB in three of the four asset categories.

The Federal Reserve sets each bank's SCB based on the bank's performance in its most recent supervisory stress test (Federal Reserve, 2020). Figure 8 shows the changes in banks' SCB requirements when the Fed introduced the SCB in 2020:Q4 and updated it in 2021:Q4.

To predict how banks respond to changes in their SCB requirements, I calculate required equity for each bank–asset pair before and after the bank's SCB requirement changes, holding other regulatory requirements and the bank's pre-event regulatory ratios fixed. I determine which capital rule binds at each SCB level, then apply the baseline supply elasticities to the changes in required equity using Proposition 1.[12]

I measure observed portfolio-share changes from the quarter before each SCB event to the same post-event horizons used for the long-run estimates. Data through 2025:Q4 provide horizons 16–20 for the 2020 event and horizon 16 for the 2021 event. I remove bank–date and asset–date–horizon fixed effects from both the observed and model-implied portfolio-share changes using the comparison sample, which includes banks whose SCB requirements stayed constant. I then average the adjusted portfolio-share changes across the same bank–event–horizon observations for banks whose SCB requirements changed.

Figure 9 shows that the model and data agree on the direction of portfolio-share changes in three of the four asset categories. The RW0 and RW0.2 portfolio shares rise in both the model and the data, and the RW1 portfolio share falls in both. The model overstates the increase in the RW0 portfolio share and understates both the increase in the RW0.2 share and the decline in the RW1 share.

[11] The baseline regressions can still use portfolio shares from quarters with SCB changes to measure responses to earlier regulatory events.

[12] The calculation retains the temporary SLR exemption in effect in 2020. For Morgan Stanley, the exemption reduces required equity on the exempt portion of RW0 holdings, so the changes in required equity need not vary linearly with risk weight.

4.4 Model-Implied Comparative Advantage

Beyond estimating asset-level own-cost elasticities of supply, I use the model to explain banks' persistent specialization in different assets through differences in the net returns that banks earn on the same asset. I use the estimated asset supply elasticities to translate cross-bank differences in portfolio shares into implied differences in net returns, after accounting for equity requirements. The estimates reveal that traditional banks have a comparative advantage in lending, while dealers and custodians have a comparative advantage in securities and reverse repos. These comparative advantages are worth several percentage points in annual net returns.

To recover bank-specific net-return levels, I assume that each asset's marginal net return declines at a constant rate across portfolio allocations. Under this assumption, a bank holding a larger share of an asset despite a higher equity requirement must have a higher net-return intercept for that asset, holding the bank's portfolio opportunity cost fixed. I specify $r_{abt}(s) = (\bar{r}_{at} + \bar{r}_{ab} + \varepsilon_{abt})s - s^2/(2\tau_a)$, where $\bar{r}_{at} + \bar{r}_{ab} + \varepsilon_{abt}$ is the intercept of marginal net returns. The common asset–date net-return level $\bar{r}_{at}$ captures variation in asset a's net-return level shared by all banks. The intercept $\bar{r}_{ab}$ is the persistent bank-specific component of the net-return level on asset a. Differences in these intercepts across banks capture comparative advantages in supplying the asset. I assume that the remaining net-return shock ε_{abt} has zero time mean for each bank–asset pair. Substituting the quadratic net-return function and return decomposition into the bank's first-order condition gives

$$\frac{s_{abt}}{\widehat{\tau}_a} + c_E k_{abt}^{Bind} = \bar{r}_{at} + \bar{r}_{ab} - \eta_{bt} + \varepsilon_{abt},$$

where η_{bt} is the bank's multiplier on the portfolio adding-up identity.

I recover the net-return intercepts $\bar{r}_{ab}$ by estimating bank–asset fixed effects in a regression of $s_{abt}/\widehat{\tau}_a + c_E k_{abt}^{Bind}$ on bank–asset, asset–date, and bank–date fixed effects. Asset–date fixed effects absorb common net-return levels, while bank–date fixed effects absorb the multiplier on the portfolio adding-up identity. I normalize the intercepts to average zero across banks within each asset and across assets within each bank. Confidence intervals use Newey–West standard errors with four quarterly lags, allowing regression errors to be correlated across observations within the same calendar quarter and across quarters up to four quarters apart. They treat the estimated supply elasticities and equity funding premium as fixed. A positive intercept indicates a comparatively high net-return level on that asset after removing bank-wide and asset-wide net-return differences.

The estimated comparative advantages align with banks' business models: Goldman Sachs and Morgan Stanley have estimated net-return intercepts of about 3 percentage points

per year in RW0 assets, which for these banks are predominantly made up of reverse repos, while custodians State Street and BNY Mellon have intercepts of about 2.5 percentage points per year in both RW0 and RW0.2 assets (Figure 10). Credit-card lenders American Express and Capital One have estimated net-return intercepts in RW1 loans of about 8 and 3 percentage points per year, respectively.

5 Application to Post-Crisis Capital Requirements

Since the financial crisis, large U.S. banks have faced stronger capital requirements while simultaneously shifting their portfolios away from lending and toward securities (Hanson et al., 2024). A prominent explanation is that stronger bank regulation has discouraged bank lending and encouraged lending by nonbanks (Buchak et al., 2018; Irani et al., 2021; Lee et al., 2024). Other work emphasizes technological change and shifts in borrower and saver demand (Buchak et al., 2024). I use the estimated asset supply elasticities to quantify the contribution of capital requirements to this shift and to the convergence in banks' portfolio compositions.

5.1 Aggregate portfolio composition

I use the estimated supply elasticities to quantify how much stronger capital requirements contributed to large banks' shift from lending toward securities. Figure 11 summarizes the post-crisis reforms. Higher buffers and bank-specific surcharges increased RBC requirements, while the new SLR became binding for banks whose portfolios were concentrated in low-risk-weight assets.

I use the actual changes in required equity for each bank, asset, and quarter to predict how much stronger capital regulation changes each asset's portfolio share at each bank. I measure these changes in required equity relative to a counterfactual with each bank's RBC requirement fixed at its 2011:Q1–2013:Q2 average and no SLR. I apply the estimated supply elasticities to these asset-level changes in required equity and account for portfolio substitution as in Proposition 1, holding bank sizes and net-return schedules fixed. I compare the model-implied and observed portfolio-share changes over the same available quarters for each bank and use fixed bank weights to aggregate the changes.

The model implies that stronger capital requirements after the financial crisis contributed to large banks' shift from lending toward securities. Figure 12 compares predicted and observed changes in large banks' portfolio composition between 2011–2013 and 2022–2024. Table 4 reports actual portfolio-share changes, model-implied changes due to capital re-

quirements, and the model-implied changes as percentages of the actual changes. The model explains about 28 percent of the shift from lending toward securities between 2011–2013 and 2022–2024, predicting an increase in securities' portfolio share minus loans' portfolio share of 3.9 percentage points compared with the observed increase of 14.0 percentage points. Table 4 reports confidence intervals for the model-implied percentage-point changes.[13]

The SLR partly offsets large banks' aggregate shift from lending to securities induced by stronger risk-based capital requirements. Higher RBC requirements raise the equity cost of high-risk-weight loans relative to securities. When the SLR becomes binding instead of the RBC, it raises required equity on low-risk-weight securities and lowers it on loans relative to the RBC requirement. The model predicts that, absent the SLR, there would have been a larger shift from lending to securities (Figure 12).

Stronger bank capital requirements therefore account for part, but not most, of large banks' shift from lending to securities over 2011–2024. Other forces, including changes in nonbank lending technology, may also have contributed (Hanson et al., 2024; Buchak et al., 2024). In general equilibrium, an aggregate strengthening of capital requirements is likely to raise loan rates as well as reduce loan quantities, so banks may shift their portfolios away from lending less than this partial-equilibrium exercise predicts. My cross-sectional estimates may also capture borrower substitution from banks facing larger capital requirement increases toward those facing smaller increases. Applying these estimates to a broad strengthening of capital requirements could therefore overstate the aggregate portfolio shift away from lending, because fewer banks remain available to absorb displaced borrowers than when requirements rise at only some banks.

5.2 Portfolio convergence across banks

Post-crisis capital requirements may also have contributed to the convergence in large banks' average asset risk weights since 2011. Since each bank tends to be bound by the rule that most strongly penalizes its core assets, risk-based capital requirements encourage lending specialists to expand securities holdings, while the leverage requirement encourages dealers and custodians to expand lending (Greenwood et al., 2017). These incentives can make banks' portfolios more similar in regulatory risk weight. I use the estimated supply elasticities to quantify how much of the observed convergence capital requirements can explain.

Consider again the two-bank example in Section 2, with Capital One specializing in credit-card loans and Goldman Sachs specializing in Treasuries. With a 10 percent RBC

[13]The model explains a larger share of the increase in securities shares (37 percent) than of the decline in loan shares (23 percent) because the observed increase in securities shares is smaller than the observed decline in loan shares, whereas the model predicts equal and opposite changes.

requirement and a 5 percent SLR, Capital One is RBC-bound and Goldman Sachs is SLR-bound. Capital One must fund each additional $100 of credit-card loans with $10 of equity, compared with $5 for Goldman Sachs. Conversely, Goldman Sachs must fund each additional $100 of Treasuries with $5 of equity, while Capital One faces no equity requirement on Treasuries. Each bank therefore faces a higher equity requirement on its specialty than the other bank does, encouraging both to expand outside their specialties.

I measure the bank's average risk weight across the four asset categories, weighted by each category's share of the assets in those categories. Convergence means that banks with higher initial average risk weights subsequently reduced them more—or increased them less—than banks with lower initial average risk weights.

Banks' ratios of risk-weighted assets to assets converged between 2011–2013 and 2022–2024, consistent with the earlier evidence in Greenwood et al. (2017). Figure 13 plots changes in RWA/assets against banks' initial RWA/assets. Goldman Sachs and Morgan Stanley began with low RWA/assets and subsequently increased their ratios. Wells Fargo, PNC, and U.S. Bancorp began with higher RWA/assets and subsequently reduced their ratios. These portfolio shifts are consistent with the incentives of SLR-bound dealers to expand lending and RBC-bound lenders to expand their securities portfolios. Regressing the observed change in RWA/assets on initial RWA/assets gives a slope of -0.14. Regressing the model-implied change on the same initial RWA/assets gives a slope of -0.05. The ratio of the model-implied slope to the observed slope implies that capital requirements account for about 35 percent of the observed convergence. Table D.1 reports confidence intervals for both slopes.

6 Conclusion

Changes in bank capital requirements typically alter equity requirements on multiple assets at once. I develop a portfolio-allocation model and empirical method to separate each asset's direct response to its own regulatory cost from portfolio substitution in response to changes in other assets' costs. In the model, each asset's own-cost elasticity governs both its direct response to its own regulatory cost and its substitution response to changes in other assets' regulatory costs. My results show that correcting for portfolio substitution is necessary to recover the magnitude and sign of each asset's own-cost elasticity of supply. I estimate positive own-cost supply elasticities for every asset category. Raising an asset's equity requirement reduces its portfolio share when other assets' equity requirements are held fixed. Securities respond more to their own regulatory costs than loans do.

Applying the estimated supply elasticities to post-crisis capital requirements, I find that stronger capital regulation accounts for about 28 percent of large banks' shift from lending toward securities since 2011 and 35 percent of the convergence in their average asset risk weights. Risk-based capital requirements encourage lending specialists to expand securities holdings, while the leverage requirement encourages dealers and custodians to expand lending. Capital regulation thus contributes to the aggregate decline in lending shares while also encouraging banks to expand outside their specialties.

These results show that the effect of capital regulation on banks' portfolio composition depends on how requirements change across the entire portfolio. Raising equity requirements on securities can encourage lending even without reducing the equity required to fund loans. Raising requirements on loans can instead encourage substitution toward securities. Evaluating a new policy therefore requires considering both its effects on the assets it directly targets and the reallocation it induces elsewhere in banks' portfolios. The framework developed here quantifies these portfolio adjustments, holding bank sizes fixed; assessing their implications for total credit provision also requires accounting for changes in bank size and lending outside the banking sector.

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Tables and Figures

Risk weight	Examples of assets
0	U.S. Treasuries; GNMA mortgage-backed securities
0.2	Federal funds; agency MBS; loans to U.S. banks
0.5	First-lien residential mortgages
1	C&I; credit cards; commercial real estate; other consumer loans

Table 1: Asset categories and standardized risk weights. [Back to text]

Notes: MBS denotes mortgage-backed securities; GNMA denotes the Government National Mortgage Association; C&I denotes commercial and industrial loans. Appendix Table B.2 lists the FR Y-9C reporting fields used to measure each asset category and the regulatory provisions assigning its risk weight. *Sources:* FR Y-9C reporting instructions; 12 CFR Part 217.

	Binding rule	C&I loans RW1	Credit cards RW1	Residential mortgages RW0.5	Agency MBS RW0.2	Treasuries RW0
G-SIBs						
BNY Mellon	SLR	5	5	5	5	5
Goldman Sachs	SLR	5	5	5	5	5
Morgan Stanley	SLR	5	5	5	5	5
State Street	SLR	5	5	5	5	5
Bank of America	RBC	12.2	12.2	6.1	2.4	0
Citigroup	RBC	13.6	13.6	6.8	2.7	0
JPMorgan Chase	RBC	13.8	13.8	6.9	2.8	0
Wells Fargo	RBC	11.3	11.3	5.7	2.3	0
Other large bank holding companies						
Capital One	RBC	11.5	11.5	5.8	2.3	0
PNC Financial	RBC	8.5	8.5	4.2	1.7	0
Truist Financial	RBC	8.8	8.8	4.4	1.8	0
U.S. Bancorp	RBC	9.1	9.1	4.6	1.8	0

Table 2: Required equity by bank and asset, 2024:Q4 (percent). [Back to text]

Notes: Required equity is the percentage of funding for each asset that must come from equity (Tier 1 capital) under the bank's binding capital rule. RBC denotes risk-based capital; SLR denotes the supplementary leverage ratio. RW denotes standardized risk weight; C&I denotes commercial and industrial loans; MBS denotes mortgage-backed securities; G-SIB denotes a global systemically important bank. For RBC-bound banks, required equity equals the required Tier 1 capital ratio times the standardized asset risk weight. For SLR-bound banks, the required equity funding share equals the bank's required SLR ratio. The bank's binding rule is the rule that requires the most Tier 1 capital given its portfolio composition and regulatory exposures. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

	No substitution correction	Pooled RW0/0.2 with substitution correction
RW0	16.65**	30.49***
	(7.36)	(10.99)
RW0.2	8.21	30.49***
	(7.40)	(10.99)
RW0.5	−12.19	11.62*
	(13.67)	(5.83)
RW1	2.50	4.41*
	(3.79)	(2.48)
N. Banks	28	28
N. Quarters	42	42
SE clustering	Bank	Bank
Adjusted R^2	0.13	0.11
Adjusted within R^2	0.00	0.00
Bank–date FE	Yes	No
Asset–date–horizon FE	Yes	Yes

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Long-run supply elasticities with and without portfolio-substitution correction. [Back to text]

Notes: RW denotes the standardized risk weight. The corrected specification imposes a common elasticity for RW0 and RW0.2 and separate elasticities for RW0.5 and RW1; the uncorrected specification estimates all four separately. Estimates pool quarters 16–20. Elasticities are recovered as $\widehat{\tau}_a = -\widehat{\beta}_a / c_E$, where $\widehat{\beta}_a$ is the portfolio-share response to required equity and $c_E = 0.05$ is the equity funding premium. Inference treats c_E as fixed. *Sources:* FR Y-9C bank regulatory filings.

	Observed change (percentage points)	Model-implied change (percentage points)	Percentage explained
RW0	4.8	1.9 [0.8, 2.9]	38.7 percent
RW0.2	0.5	0.1 [-0.7, 0.9]	21.5 percent
RW0.5	-2.3	-1.0 [-1.8, -0.1]	42.4 percent
RW1	-6.4	-1.0 [-2.1, 0.1]	15.6 percent
Securities minus loans	14.0	3.9 [1.1, 6.8]	28.2 percent

Table 4: Observed and model-implied changes in aggregate portfolio shares. [Back to text]

Notes: Changes compare 2011:Q1–2013:Q2 with 2022–2024. Bank weights are fixed at pre-period averages. Observed and model-implied changes use the same available quarters within each bank and comparison period. Predictions hold bank sizes and net-return schedules fixed. Brackets report 95 percent confidence intervals reflecting uncertainty in the bank-clustered supply-elasticity estimates. The final row subtracts the change in RW0.5 and RW1 shares from the change in RW0 and RW0.2 shares; its interval accounts jointly for estimation uncertainty across all four categories. Percentages are point estimates only. Shares exclude trading assets, reserves, and vault cash. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

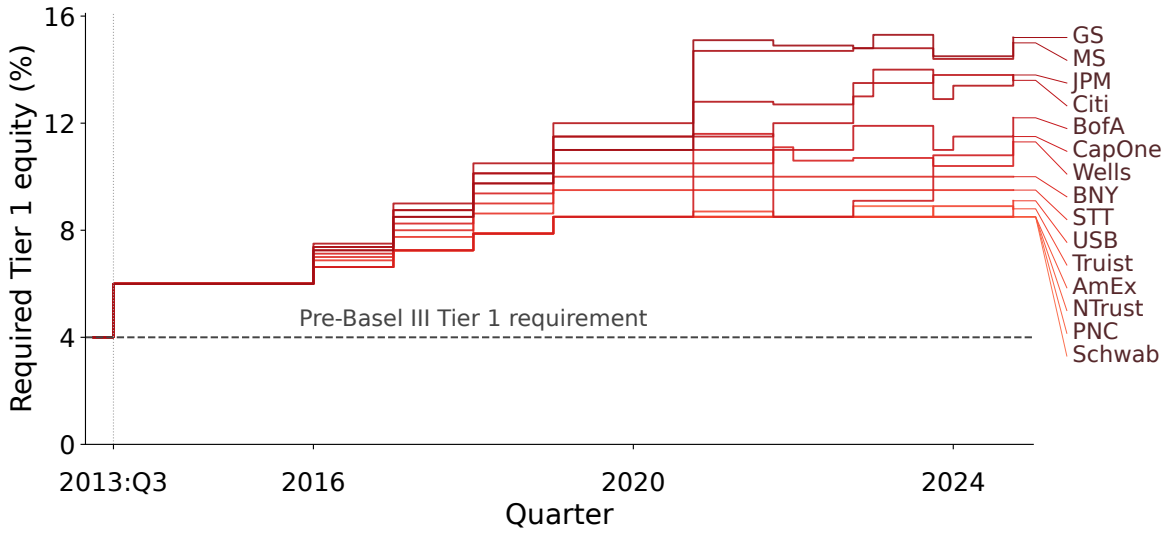


Figure 1: Bank-level RBC requirements over time. [Back to text]

Notes: Each line shows a bank's required Tier 1 capital as a percentage of risk-weighted assets, including the minimum ratio, buffers, and surcharges. The dashed horizontal line marks the pre-Basel III minimum. RBC denotes risk-based capital. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules.

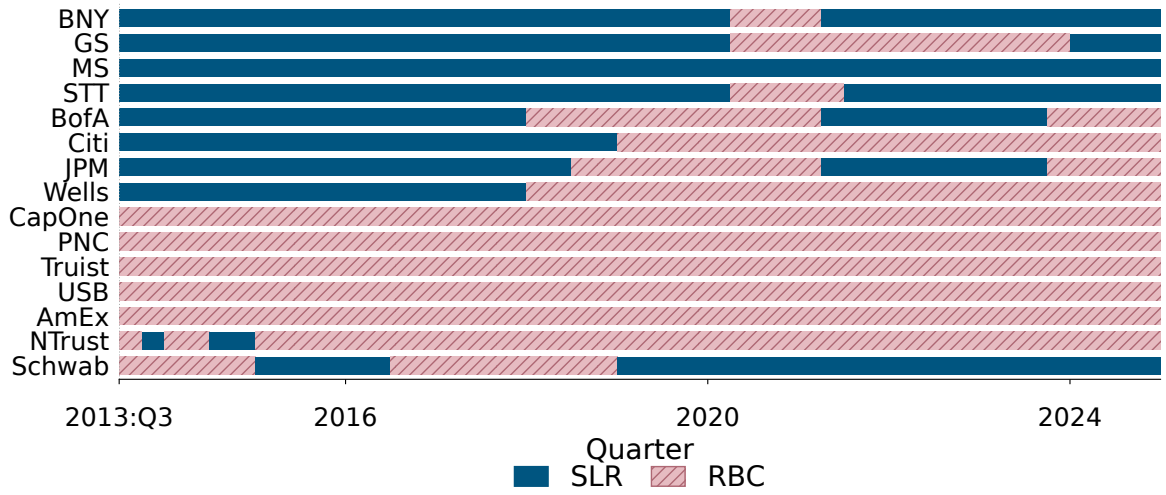


Figure 2: Binding capital rule by bank. [Back to text]

Notes: Each row represents a bank. Hatched light-red segments indicate a binding risk-based capital (RBC) requirement; solid blue segments indicate a binding supplementary leverage ratio (SLR) requirement. The bank's binding rule is the rule that requires the most Tier 1 capital given its portfolio composition and regulatory exposures. The figure identifies the binding rule by comparing RBC and SLR requirements, excluding the separate Tier 1 leverage requirement. The estimation includes the separate Tier 1 leverage requirement, which rarely binds. Before 2016:Q3, I approximate the denominator of the SLR using the method described in Appendix B. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

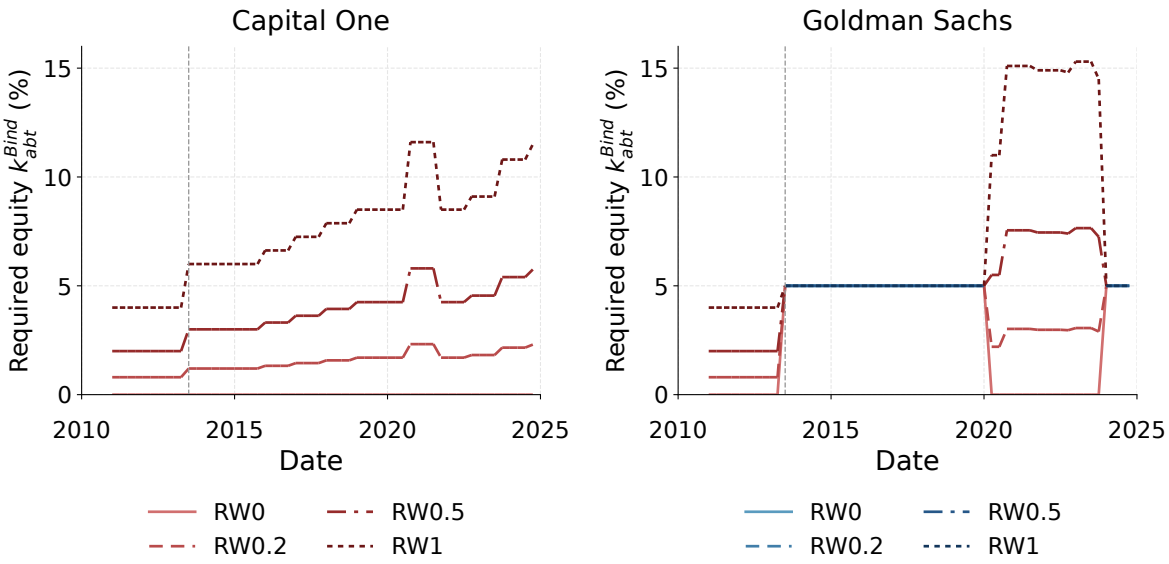


Figure 3: Required equity by asset over time: Capital One and Goldman Sachs. [Back to text]

Notes: Lines show the percentage of funding for assets in each risk-weight (RW) category that must come from equity (Tier 1 capital) under the bank's binding capital rule. Under a binding risk-based capital requirement, required equity equals the bank-level required ratio times RW. Under a binding supplementary leverage ratio (SLR), the required equity funding share for a nonexempt asset equals the bank's required SLR ratio. The bank's binding rule is the rule that requires the most Tier 1 capital given its portfolio composition and regulatory exposures. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

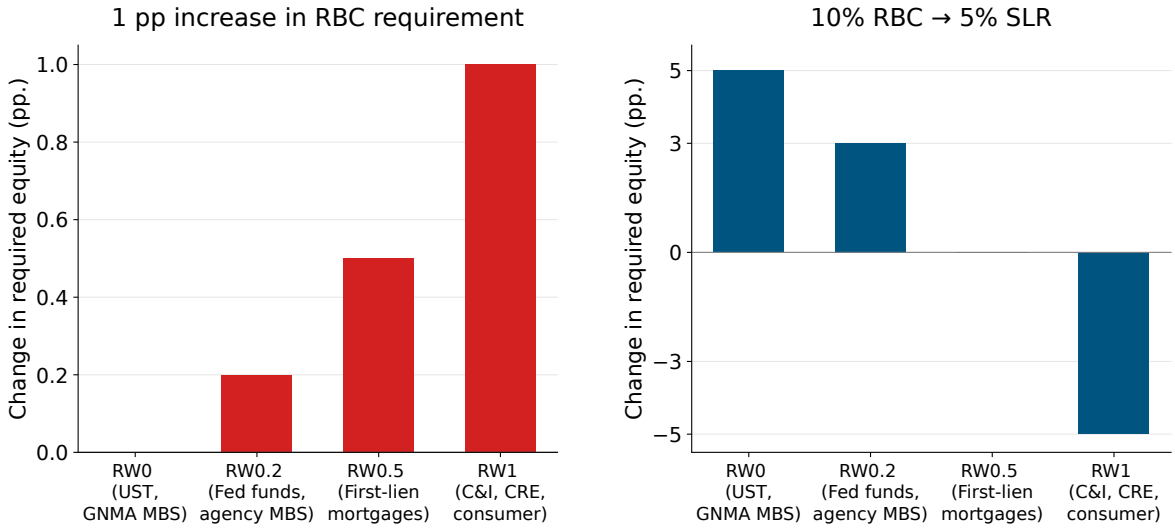


Figure 4: Changes in required equity across assets under an RBC increase and an RBC-to-SLR switch. [Back to text]

Notes: Bars show how required equity in each risk-weight (RW) category changes under two hypothetical regulatory changes, in percentage points. The left panel raises a binding risk-based capital (RBC) requirement by one percentage point, assuming RBC remains binding. The right panel replaces a binding 10 percent RBC requirement with a binding 5 percent supplementary leverage ratio (SLR) requirement.

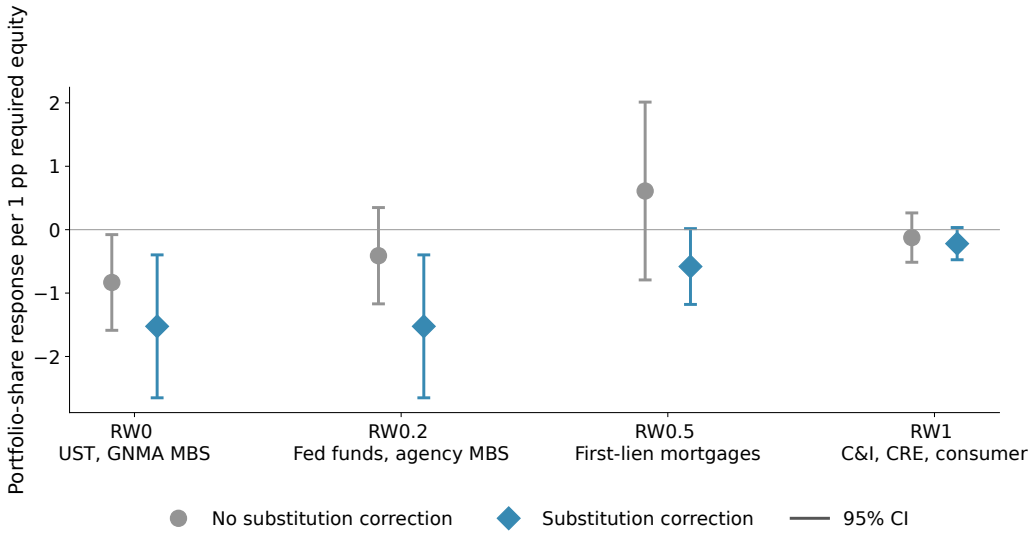


Figure 5: Long-run portfolio-share responses to higher required equity. [Back to text]

Notes: Points show portfolio-share responses (percentage points) to a one-percentage-point increase in required equity, pooling quarters 16–20. Gray circles omit substitution correction; blue diamonds include it and impose a common elasticity for RW0 and RW0.2. Error bars show 95 percent bank-clustered confidence intervals. Estimates use portfolio shares for 28 banks and include asset–date–horizon fixed effects. Portfolio shares exclude trading assets, reserves, and vault cash from the denominator; the uncorrected specification also includes bank–date fixed effects. RW denotes risk weight. Table 3 reports the own-cost supply elasticities and standard errors implied by the response coefficients. *Sources:* FR Y-9C bank regulatory filings.

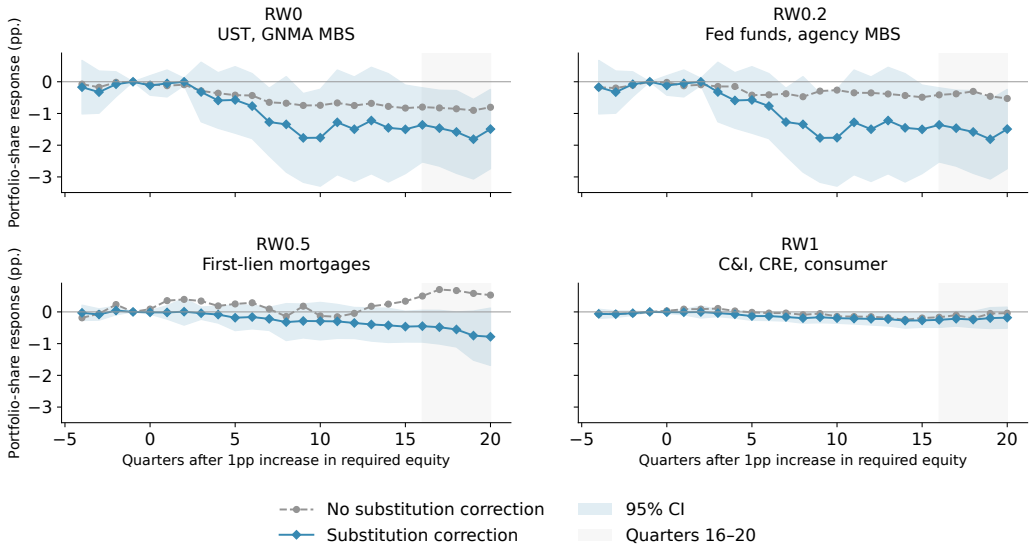


Figure 6: Dynamic asset-share responses with and without portfolio-substitution correction. [Back to text]

Notes: Lines show portfolio-share responses, in percentage points, to a one-percentage-point increase in required equity. Dashed gray lines with circles show estimates without substitution correction; solid blue lines with diamonds show estimates with portfolio-substitution correction that impose a common supply elasticity for RW0 and RW0.2 assets. RW denotes risk weight. Blue shading gives 95 percent confidence intervals clustered by bank. Quarter −1 is the zero reference point; quarter 0 includes the event-quarter response. Light-gray shading marks quarters 16–20, pooled in the long-run estimates. Estimates use portfolio shares for 28 banks, excluding trading assets, reserves, and vault cash from the denominator, and include asset–date–horizon fixed effects. The uncorrected specification also includes bank–date fixed effects. *Sources:* FR Y-9C bank regulatory filings.

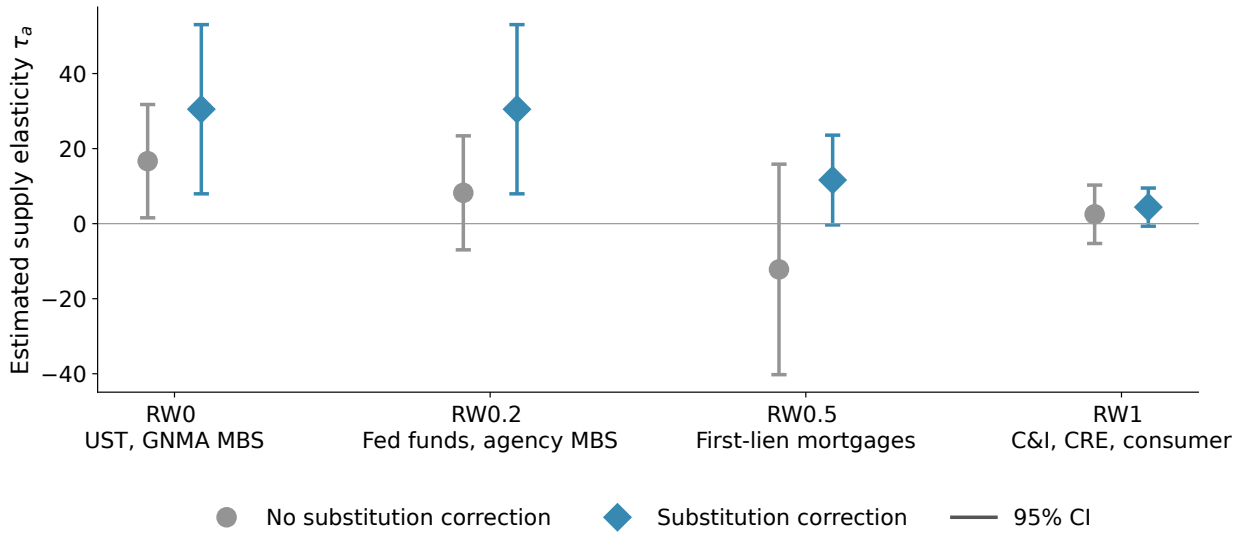


Figure 7: Long-run implied asset-supply elasticities. [Back to text]

Notes: Points show supply elasticities, $\widehat{\tau}_a = -\widehat{\beta}_a/c_E$, where $\widehat{\beta}_a$ is the long-run portfolio-share response to required equity and $c_E = 0.05$ is the equity funding premium. Gray circles omit substitution correction; blue diamonds include the correction and impose a common supply elasticity for RW0 and RW0.2 assets. RW denotes risk weight. Error bars give 95 percent confidence intervals clustered by bank, treating c_E as fixed. Table 3 reports the baseline own-cost supply elasticity estimates and standard errors. Estimates use portfolio shares for 28 banks, excluding trading assets, reserves, and vault cash from the denominator, and include asset–date–horizon fixed effects. The uncorrected specification also includes bank–date fixed effects. Long-run estimates pool quarters 16–20 after the change in required equity. *Sources:* FR Y-9C bank regulatory filings.

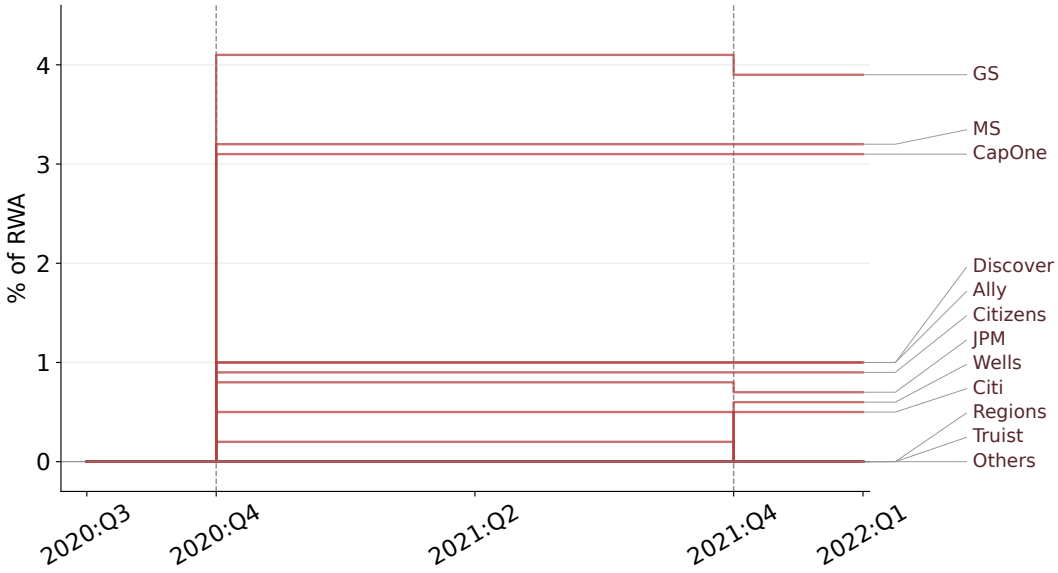


Figure 8: Stress Capital Buffer by bank in 2020 and 2021. [Back to text]

Notes: Lines cumulate each bank's Stress Capital Buffer changes in 2020 and 2021, in percentage points of risk-weighted assets (RWA). Bank abbreviations label the lines at the right; Others denotes banks with no buffer change. Dashed vertical lines mark 2020:Q4 and 2021:Q4, when the new buffers took effect. *Sources:* Federal Reserve capital-requirement announcements for 2020 and 2021.

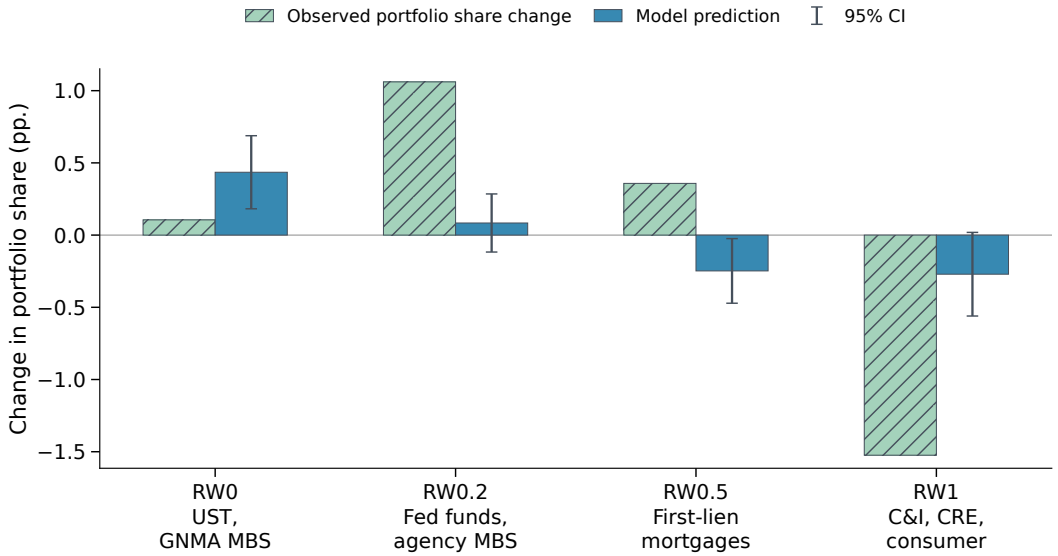


Figure 9: Predicted and observed portfolio-share responses to SCB changes. [Back to text]

Notes: Blue bars (Model prediction) apply the baseline supply elasticities estimated in Section 4.1 to changes in required equity caused by Stress Capital Buffer (SCB) changes, incorporating portfolio substitution. Hatched green bars (Observed portfolio share change) show observed portfolio-share changes from the quarter before each SCB event to available quarters between 16 and 20 afterward. Both predicted and observed portfolio-share changes are residualized with respect to bank–date and asset–date–horizon fixed effects using all available observations, including observations for banks whose SCB requirements did not change. Bars average the residualized portfolio-share changes following 15 changes in banks' SCB requirements at 11 banks in 2020 and 2021. Bars are in percentage points. Error bars on model predictions show 95 percent delta-method confidence intervals using bank-clustered uncertainty in the baseline supply-elasticity estimates. Portfolio shares exclude trading assets, reserves, and vault cash from the denominator. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve capital-requirement announcements for 2020 and 2021.

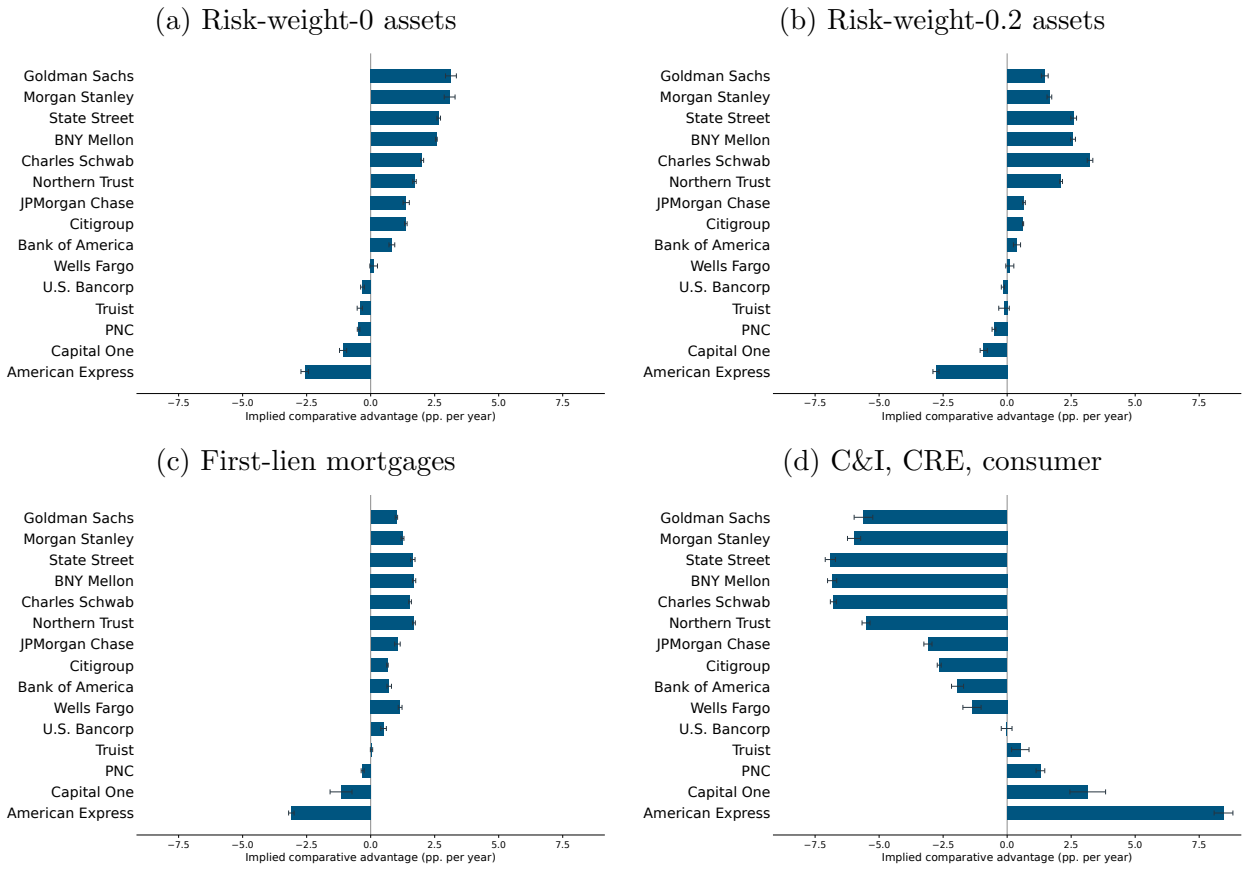


Figure 10: Implied comparative advantage across asset categories. [Back to text]

Notes: Bars show estimated bank-specific intercepts in marginal net returns, in percentage points per year, after accounting for equity requirements. The intercepts are normalized to average zero across banks within each asset and across assets within each bank. Error bars show 95 percent confidence intervals allowing for serial correlation across quarters, treating the equity funding premium and estimated supply elasticities as fixed. *Sources:* FR Y-9C bank regulatory filings.

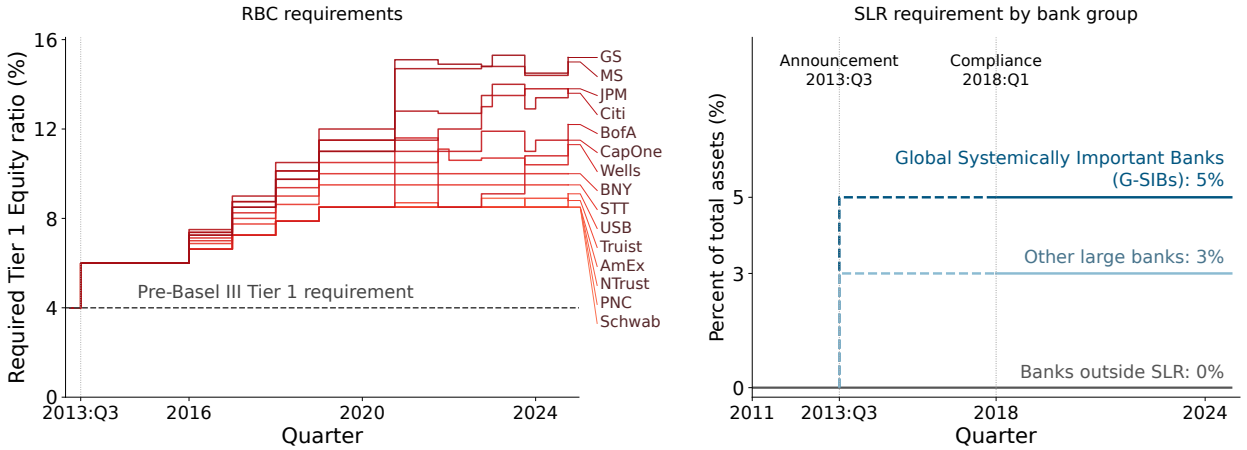


Figure 11: Historical bank-level RBC and SLR requirements. [Back to text]

Notes: The left panel shows each bank's required Tier 1 capital as a percentage of risk-weighted assets under risk-based capital (RBC) requirements. The right panel shows supplementary leverage ratio (SLR) requirements by bank group; G-SIB denotes a global systemically important bank. Vertical lines mark the SLR announcement and compliance dates. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

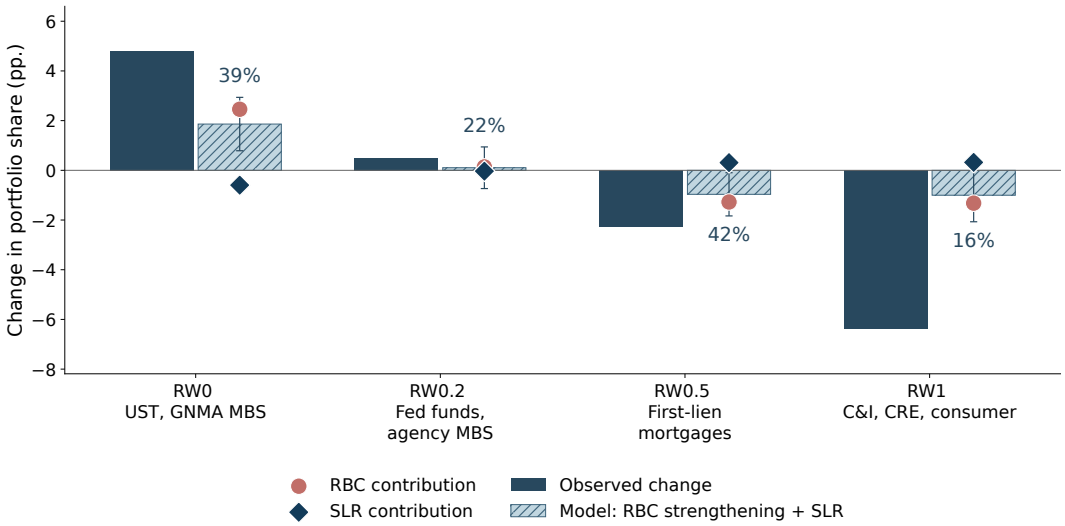


Figure 12: Observed aggregate portfolio shifts and predicted effects of RBC strengthening and SLR introduction. [Back to text]

Notes: Solid bars show observed portfolio-share changes from 2011:Q1–2013:Q2 to 2022–2024; hatched bars show predictions under actual requirements relative to pre-period RBC requirements without the SLR. Red circles show the RBC contribution; blue diamonds show the additional SLR contribution. Labels give the percentage of the observed change explained. Shares exclude trading assets, reserves, and vault cash. Bank weights and portfolio sizes are fixed at pre-period averages; observed and predicted changes use the same available quarters within each bank and comparison period. Error bars give 95 percent delta-method confidence intervals. RW denotes risk weight; RBC denotes risk-based capital; SLR denotes supplementary leverage ratio. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

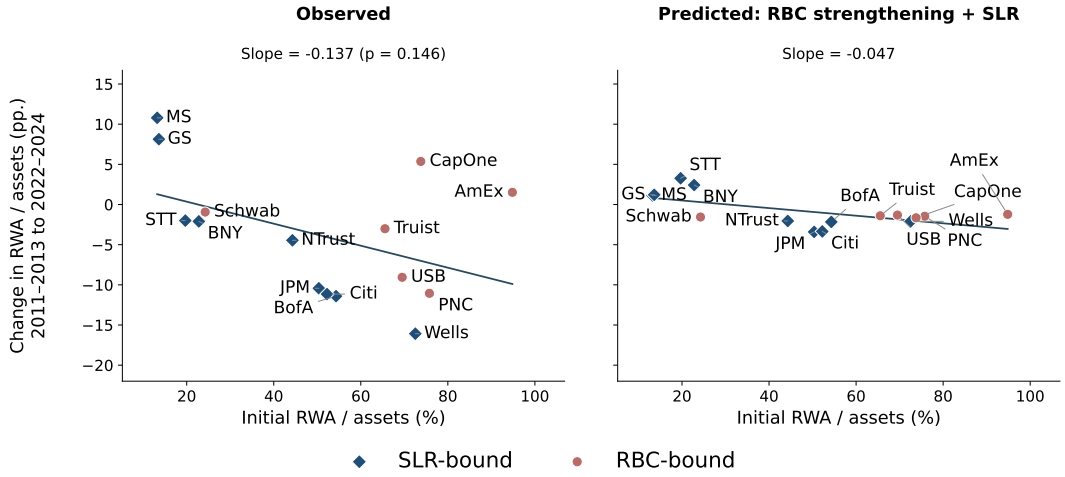


Figure 13: Observed and predicted convergence in banks' ratios of risk-weighted assets to assets. [Back to text]

Notes: Each point is a bank. RWA/assets is the average risk weight across the four asset categories, weighted by the dollar amount of assets in each category and excluding reserves and vault cash. Portfolio changes compare the 2011:Q1–2013:Q2 and 2022–2024 averages. Predictions hold each bank's total assets in the four categories fixed. Blue diamonds identify SLR-bound banks and red circles RBC-bound banks in 2013:Q4. Fitted lines weight banks equally; the observed-slope p-value uses heteroskedasticity-robust standard errors. RWA denotes risk-weighted assets; RBC denotes risk-based capital; SLR denotes supplementary leverage ratio. *Sources:* FR Y-9C bank regulatory filings; Federal Reserve Basel III (2013), G-SIB surcharge (2015), and stress capital buffer (2020) rules; enhanced SLR rule (2014).

Appendix

A Model derivations and extensions

A.1 Proof of Proposition 1

Back

Indices a, b, and t denote assets, banks, and quarters. The portfolio share s_{abt} and equity ratio E_{bt} are measured relative to bank b's total assets. The symbols ℓ_{bt}^{RBC} and ℓ_{bt}^{SLR} denote required bank-level capital ratios, and w_a is asset a's risk weight. The equity funding premium is c_E, the supply elasticity is τ_a, and η_{bt} is the portfolio-constraint multiplier. The operator Δ denotes a change from quarter $t-1$ to quarter t.

For an interior portfolio, let λ_{bt}^{RBC} and λ_{bt}^{SLR} be the capital-constraint multipliers. The first-order condition for equity and complementary slackness for the RBC and SLR constraints give

$$\lambda_{bt}^{RBC} + \lambda_{bt}^{SLR} = c_E, \quad \lambda_{bt}^{RBC}\left(E_{bt} - \ell_{bt}^{RBC}\sum_a w_a s_{abt}\right) = 0, \quad \lambda_{bt}^{SLR}\left(E_{bt} - \ell_{bt}^{SLR}\sum_a s_{abt}\right) = 0.$$

When the RBC and SLR require different amounts of Tier 1 capital, the slack rule has multiplier zero and the binding rule has multiplier c_E. Differentiating the share choice in the constrained problem therefore gives the asset first-order condition in equation (3), with marginal required equity defined in equation (2).

The second-order approximation to asset a's net-return function r_{abt} around the fixed reference portfolio share $\bar{s}_{ab}$ is

$$r_{abt}(s) \approx r_{abt}(\bar{s}_{ab}) + r'_{abt}(\bar{s}_{ab})(s - \bar{s}_{ab}) - \frac{(s-\bar{s}_{ab})^2}{2\tau_a}.$$

Differentiating this displayed quadratic polynomial with respect to s gives

$$r'_{abt}(s) \approx r'_{abt}(\bar{s}_{ab}) - \frac{s - \bar{s}_{ab}}{\tau_a}.$$

Taking differences between dates $t-1$ and t, while holding the reference share $\bar{s}_{ab}$ fixed, gives $\Delta r\prime_{abt}(\bar{s}_{ab}) - \Delta s_{abt}/\tau_a$ for the change in marginal net returns along the chosen portfolios. Subtracting the asset first-order condition (3) at date $t-1$ from the condition at date t then yields

$$\Delta s_{abt} \approx \tau_a\big(\Delta r'_{abt}(\bar{s}_{ab}) - c_E \Delta k_{abt}^{Bind} - \Delta\eta_{bt}\big).$$

Holding net-return schedules fixed and summing across assets, the fixed portfolio total implies

$$0 = \sum_a \Delta s_{abt} \approx -c_E \sum_a \tau_a \Delta k_{abt}^{Bind} - \Delta\eta_{bt} \sum_a \tau_a, \qquad \Delta\eta_{bt} \approx -c_E \frac{\sum_a \tau_a \Delta k_{abt}^{Bind}}{\sum_a \tau_a}.$$

Substituting the expression for the change in the portfolio-constraint multiplier, $\Delta\eta_{bt}$, into equation (4), with net-return schedules held fixed, gives the portfolio-share response in equation (5). Approximation errors vanish to first order near the reference portfolio. Globally quadratic net returns make the marginal-net-return approximation exact.

A.2 From marginal net returns to the estimating equation

Back

Allowing marginal net returns to change, summing the asset-share first-order conditions and imposing the portfolio adding-up identity $\sum_a \Delta s_{abt} = 0$ gives

$$\Delta s_{abt} \approx \tau_a \left[\Delta r'_{abt}(\bar{s}_{ab}) - \frac{\sum_a \tau_a \Delta r'_{abt}(\bar{s}_{ab})}{\sum_a \tau_a} - c_E \left(\Delta k_{abt}^{Bind} - \Delta \bar{k}_{bt}^{\tau} \right) \right]. \tag{7}$$

Assume that changes in marginal net returns at fixed reference shares satisfy

$$\Delta r'_{abt}(\bar{s}_{ab}) = \Delta \bar{r}_{at} + \Delta \bar{r}_{bt} + \Delta \varepsilon_{abt}.$$

The component $\Delta\bar{r}_{at}$ is common across banks holding asset a, $\Delta\bar{r}_{bt}$ is common across assets held by bank b, and $\Delta\varepsilon_{abt}$ is a bank–asset–date return shock. I assume that $\Delta\varepsilon_{abt}$ is uncorrelated with regulatory changes in equity requirements across the bank's assets, conditional on asset–date effects. Substituting this decomposition into equation (7) gives

$$\begin{aligned} \Delta s_{abt} \approx & \underbrace{\tau_a \left[\Delta \bar{r}_{at} - \frac{\sum_a \tau_a \Delta \bar{r}_{at}}{\sum_a \tau_a} \right]}_{\delta_{at}} + \underbrace{\tau_a \left[\Delta \bar{r}_{bt} - \frac{\sum_a \tau_a \Delta \bar{r}_{bt}}{\sum_a \tau_a} \right]}_{=0} \\ & - c_E \tau_a \left(\Delta k_{abt}^{Bind} - \Delta \bar{k}_{bt}^{\tau} \right) + \underbrace{\tau_a \left[\Delta \varepsilon_{abt} - \frac{\sum_a \tau_a \Delta \varepsilon_{abt}}{\sum_a \tau_a} \right]}_{u_{abt}}. \end{aligned}$$

The underbraced term δ_{at} is the asset–date component of the portfolio-share response; u_{abt} is the remaining bank–asset–date component of the portfolio-share response. The bank–date return component $\Delta\bar{r}_{bt}$ is the same for every asset held by bank b. Its supply-elasticity-weighted average across assets, $\sum_a \tau_a \Delta\bar{r}_{bt} / \sum_a \tau_a$, therefore equals $\Delta\bar{r}_{bt}$ itself, so the bank–date component cancels from the portfolio-share response. With supply elasticities common

across banks, δ_{at} depends only on asset and date and is absorbed by asset–date fixed effects. After canceling the bank–date return component, the portfolio-share response becomes

$$\Delta s_{abt} \approx \delta_{at} - c_E \tau_a \left(\Delta k_{abt}^{Bind} - \Delta \bar{k}_{bt}^{\tau} \right) + u_{abt}.$$

The model's portfolio-share response yields the local projection in equation (6).

A.3 Adjustment costs and long-run portfolio responses

Back

Portfolio adjustment costs can delay banks' responses to required equity without changing their long-run portfolio choices. Let $\mathbf{s}_{bt}$ be the vector of bank b's asset shares, and define the portfolio change as $\Delta \mathbf{s}_{bt} = \mathbf{s}_{bt} - \mathbf{s}_{b,t-1}$. The adjustment-cost function $G_b(\Delta \mathbf{s}_{bt})$ measures the cost of that reallocation per dollar of total assets. I assume that the adjustment-cost function is convex and continuously differentiable, with $G_b(\mathbf{0}) = 0$ and $\nabla G_b(\mathbf{0}) = \mathbf{0}$, where ∇G_b is the vector of marginal adjustment costs. Thus, leaving the portfolio unchanged incurs no adjustment cost, and the marginal adjustment cost is zero at an unchanged portfolio. The function can differ across banks and assets and depend jointly on changes in all asset shares; I do not impose a quadratic or other parametric form.

The bank maximizes the discounted sum of period profits from Section 2, subtracting the adjustment cost $G_b(\Delta \mathbf{s}_{bt})$ in each period, subject to the RBC and SLR capital constraints and a fixed portfolio total in each period. Let $0 < \rho < 1$ be its discount factor, and write $G_{b,a}$ for the partial derivative of the adjustment-cost function with respect to asset a's portfolio share change. At an interior optimum with one binding capital rule, the asset-share first-order condition is

$$r'_{abt}(s_{abt}) - c_E k_{abt}^{Bind} - \eta_{bt} = G_{b,a}(\Delta \mathbf{s}_{bt}) - \rho \mathbb{E}_t[G_{b,a}(\Delta \mathbf{s}_{b,t+1})]. \tag{8}$$

Here $\mathbb{E}_t$ denotes expectation conditional on information in quarter t. Increasing asset a's current portfolio share incurs a marginal adjustment cost today and reduces the portfolio-share adjustment needed to reach a given allocation next period. The right-hand side of equation (8) is the current marginal adjustment cost minus the discounted expected marginal adjustment cost saved next period.

Consider an unanticipated, permanent change in capital requirements from an initial stationary portfolio, holding net-return schedules and total portfolio size fixed. Suppose the optimal portfolio converges to an interior stationary portfolio and the expected marginal adjustment cost converges to zero. Continuity of ∇G_b and $\nabla G_b(\mathbf{0}) = \mathbf{0}$ imply that the

right-hand side of equation (8) vanishes as adjustment ends. Once adjustment costs vanish, equation (8) reduces to the static asset first-order condition (3) used in Proposition 1. Linearizing marginal net returns around the initial portfolio gives the long-run response

$$\Delta s_{ab}^{LR} \approx -c_E \tau_a \left(\Delta k_{ab}^{Bind} - \frac{\sum_{a'} \tau_{a'} \Delta k_{a'b}^{Bind}}{\sum_{a'} \tau_{a'}} \right).$$

Here Δs_{ab}^{LR} is the change in asset a's share between the initial and final stationary portfolios, and Δk_{ab}^{Bind} is the corresponding change in its binding equity requirement. Thus, a local projection at a horizon when adjustment is substantially complete recovers the static supply response without specifying the shape of adjustment costs. At shorter horizons, its coefficient also reflects the marginal adjustment costs in equation (8). The dynamic portfolio-share estimates in Figure 6 indicate whether the response has stabilized by quarters 16–20, the horizons pooled for long-run estimation.

B Data construction

B.1 Mergers

Back

Acquirer	Target	Close	Asset jump at close (percent)
BB&T / Truist	SunTrust	2019:Q4	100
M&T	Hudson City	2015:Q4	26
M&T	People's United	2022:Q2	36
First Citizens	First Citizens Bcp	2014:Q4	37
First Citizens	CIT	2022:Q1	86
First Citizens	SVB	2023:Q1	96
BMO	M&I	2011:Q3	57
BMO	Bank of the West	2023:Q1	44
Huntington	FirstMerit	2016:Q3	36
Huntington	TCF Financial	2021:Q2	39
KeyCorp	First Niagara	2016:Q3	34
Charles Schwab	TD Ameritrade	2020:Q4	31
NYCB / Flagstar	Flagstar	2022:Q4	43
NYCB / Flagstar	Signature (deposits)	2023:Q1	37
Capital One	ING Direct USA	2012:Q1	43
Capital One	Discover	2025:Q2	34

Table B.1: Acquisitions triggering a bank-entity split [Back to text]

Notes: This table lists acquisitions for which banks are split into separate entities when the target exceeds 25 percent of the acquirer's assets at close. Asset jump at close is the acquirer's change in total assets from the preceding quarter to the closing quarter, as a percentage of preceding-quarter assets (FR Y-9C item BHCK2170). It measures the balance-sheet discontinuity removed by the split, not the target-to-acquirer size ratio. *Sources:* FR Y-9C bank regulatory filings (asset amounts).

B.2 Risk-weight mapping

Back

Asset	FR Y-9C mnemonic	w_a	12 CFR 217
Risk weight 0			
U.S. Treasury securities	BHCK0211, 1286	0	§32(a)(1)(i)
GNMA pass-through MBS	BHCKG300, G302	0	§32(a)(1)(i)
Risk weight 0.2			
Agency and GSE MBS and CMOs	BHCKG304, G306, G312, G314, G316, G318, K142, K144, K150, K152	0.2	§32(c)(1), §43
Agency and GSE non-MBS debt	BHCK1289, 1291, 1294, 1297, HT50, HT52	0.2	§32(c)(1)
Federal funds sold	BHDMB987	0.2	§32(d)(1)
Loans to U.S. depository institutions	BHCK1292	0.2	§32(d)(1)
Risk weight 0.5			
First-lien closed-end 1–4 family residential mortgages	BHDM5367	0.5	§32(g)(1)
Risk weight 1			
Commercial and industrial loans	BHCK1763, 1764, KX56	1	§32(f)(1)
Nonfarm nonresidential real-estate loans	BHCKF160, F161	1	§32(f)(1)
Junior-lien residential mortgages and home-equity lines	BHDM1797, 5368	1	§32(g)(2)
American Express charge-card reporting component	BHCKJ451 (Amex only)	1	§32(l)(5)
Credit-card loans	BHCKB538	1	§32(l)(5)
Other consumer loans	BHCKB539, K137, K207	1	§32(l)(5)
Agricultural loans	BHCK1590	1	§32(f)(1), (l)(5)
Finance lease receivables	BHDM2165	1	§32(l)(5)
Split across the classes by the bank's reported allocation			
Cash and depository balances (excluding the zero-weight portion)	BHCK0081, 0395	0.2, 0.5, 1	§32(d)
Securities purchased under agreements to resell	BHCKB989	0, 0.2, 1	§37

Table B.2: Asset reporting fields, standardized risk weights, and applicable regulatory provisions [Back to text]

Notes: FR Y-9C denotes the Consolidated Financial Statements for Holding Companies. Reporting-field identifiers specify the asset amounts used to construct each category; securities amounts are measured at amortized cost. The symbol w_a denotes asset a's standardized risk weight. GNMA denotes the Government National Mortgage Association; MBS denotes mortgage-backed securities; GSE denotes government-sponsored enterprise; CMO denotes collateralized mortgage obligation. *Sources:* FR Y-9C reporting instructions; 12 CFR Part 217.

Securities are measured at amortized cost, combining held-to-maturity and available-for-sale holdings. Legal references in Table B.2 refer to 12 CFR Part 217.

Mixed-weight categories. Agency non-MBS debt combines full-faith-and-credit obligations with GSE obligations, reported jointly from 2018:Q2; the category is assigned a

weight of 0.2 throughout. Federal funds sold are assigned 0.2. Private-label CMOs backed by agency collateral are assigned the 0.2 securitization floor (§217.43). First-lien residential mortgages are assigned 0.5, although Schedule HC-C does not identify all underwriting and performance conditions required for that weight.

Reverse repos. The 2015 revision to Schedule HC-R removed the risk-weight breakdown used to allocate reverse repos across asset categories. Before 2015, the reported breakdown combined reverse repos and federal funds sold. I use this breakdown to approximate the reverse-repo allocation. To construct a consistent series across the reporting change, I apply each bank's mean risk-weight fractions from 2011–2014 to its reverse-repo holdings in every quarter. For banks without usable historical reports, I use the sample's repo-weighted allocation. Holding these fractions fixed prevents the change in reporting from generating portfolio reallocation in the measured asset shares. The constructed reverse-repo holdings by risk-weight category capture changes in total reverse-repo holdings while holding the risk-weight composition of reverse repos fixed. The regulatory treatment depends on collateral and counterparty characteristics (§217.37).

Cash and trading assets. I exclude Federal Reserve balances, vault cash, and all trading assets from both modeled asset holdings and the total assets used to calculate portfolio shares. Trading Treasuries have a zero specific-risk weight and are subject to the general market-risk capital requirement assessed on the trading portfolio (§§217.205, 217.210).

Loan coverage. Loans to nondepository financial institutions (BHCKJ454) and loans for purchasing or carrying securities (BHCK1545) enter the share denominator through total loans (BHCK2122) but are not assigned to a modeled risk-weight category. Other loans reported in BHCKJ451 also remain outside the modeled categories, with an exception for the American Express reporting correction described in Appendix B.

Cash and depository balances. Cash and balances at depository institutions can receive different risk weights. I use the fractions reported in Schedule HC-R for each bank and quarter to allocate its cash and U.S. depository balances (BHCK0081 and BHCK0395) across risk-weight categories. When a quarter's allocation is missing, I use the bank's mean fractions over quarters with reported allocations. If the bank has no usable reported allocation, I use the average fractions across reporting banks and quarters, weighted by cash and U.S. depository balances. I exclude the zero-weight portion from modeled holdings and from the total assets used to calculate portfolio shares. The portions assigned weights of 0.2, 0.5, and 1 enter the corresponding modeled categories. The portion assigned a weight of 1.5 remains in the total assets used to calculate portfolio shares but outside the modeled categories.

B.3 Data cleaning

Back

American Express. In 2016:Q3, American Express's charge-card receivables moved from HC-C item 9.b.(2), BHCKJ451 ("All other loans, excluding consumer loans"), into loan categories already included in risk weight 1. I include Amex's J451 holdings in risk weight 1 throughout the sample as a separate component to maintain consistent coverage across this reporting change. Including these holdings reduces the apparent discontinuity in C&I and other consumer loans from approximately $45.6 billion to $0.36 billion. Total loans already include these holdings, so the correction leaves the total assets used to calculate portfolio shares unchanged. The Federal Reserve's June 2017 FR Y-14 reporting notice documents updated guidance moving consumer charge cards from item 9.b.(2) to item 6.d; the corporate component of the reclassification is inferred from the reported balances. J451 holdings at other banks remain outside the modeled risk-weight categories.

Discover. Discover reported $10.7 billion of borrowed mortgage-backed securities in 2020:Q3 from a temporary securities-lending transaction that matured in November 2020. I subtract this amount from Discover's agency MBS holdings and from the total assets used to calculate its portfolio shares in 2020:Q3. The amount is the rounded carrying value and fair value of the collateral reported in Discover's 2020:Q3 Form 10-Q, Note 2, Investments.

Binding-rule smoothing. I smooth the quarterly binding-rule classifications using three adjustments in sequence. First, if a bank is classified under rule A, then rule B for two quarters, and then a different rule C, I assign those two quarters to rule C: A–B–B–C becomes A–C–C–C. Second, when a bank returns to its previous rule after at most four quarters under another rule, I assign the previous rule to those intervening quarters and repeat this step until no further changes occur. Spells lasting at least three quarters that begin in 2020:Q2–2021:Q1 are preserved. Third, I remove one- or two-quarter deviations between spells of the same rule that each last at least three quarters; the intervening quarters may be classified under different rules. These adjustments use both preceding and subsequent classifications.

B.4 Estimating banks' SLRs before reporting began

Back

Leverage exposure. Banks began reporting the SLR on the FR Y-9C in 2016:Q3. For 2013:Q1–2016:Q2, I construct a proxy by dividing reported Tier 1 capital by estimated total leverage exposure. Estimated exposure is the sum of on-balance-sheet assets used for the leverage ratio, credit-equivalent derivatives exposures, other off-balance-sheet credit-equivalent exposures, repo-style counterparty exposures, and an allowance for commitments

omitted from the risk-based exposure measure. Tables B.3 and B.4 list the reporting items and their construction. Missing component amounts are treated as zero when summed. The proxy is available for banks with reported SLRs in the calibration period.

Repo-style exposure. Before 2015, I impute repo-style counterparty exposure by multiplying gross repo-style balances by the bank's mean ratio of BHCKS516 to those balances over 2015:Q1–2016:Q2. Gross balances include federal funds sold and purchased, reverse repos, and repos. If the bank lacks a usable ratio, I use BHCE3433. From 2015:Q1 onward, I use BHCKS516 directly. In 224 bank-quarters over 2016:Q3–2019:Q4, the imputed and reported exposures have a correlation of 0.967; aggregate imputed exposure divided by aggregate reported exposure is 0.881.

Reporting changes. The derivatives exposure definition changes in 2015:Q1. I rescale the pre-2015 derivatives series by the ratio of mean derivative exposure relative to on-balance-sheet assets in 2015:Q1–Q2 to the corresponding mean in 2014:Q3–Q4. I also check new-format off-balance-sheet items for reporting breaks during 2015:Q2–2016:Q2. A break must exceed 5 percent of the preceding quarter's constructed leverage exposure both in the quarterly change and in the difference between the available four-quarter means before and after the break. This procedure identifies Morgan Stanley's BHCKG619 in 2016:Q1. For Morgan Stanley's earlier quarters under the revised reporting format, I replace BHCKG619 with its mean ratio to on-balance-sheet assets in 2016:Q1–Q2, multiplied by contemporaneous on-balance-sheet assets.

Calibration. I infer reported leverage exposure from Tier 1 capital divided by the reported SLR. For each bank, I average the ratio of reported to constructed exposure over 2016:Q3–2019:Q4 and multiply constructed exposure by this factor in every quarter. The proxy omits a separate adjustment for written credit derivatives because purchased protection needed to calculate the net exposure is unavailable in the extract. The bank-specific ratio of reported to constructed leverage exposure adjusts the proxy for average differences between the two exposure measures. I assess fit over the calibration period and in a held-out comparison that estimates the factor over 2016:Q3–2017:Q4 and evaluates it over 2018:Q1–2019:Q4.

FR Y-9C inputs. Table B.3 identifies the FR Y-9C inputs used to construct the SLR proxy. Dollar amounts are in the FR Y-9C reporting units; the proxy SLR is 100 times Tier 1 capital divided by calibrated leverage exposure.

Commitments. The commitment adjustment uses Schedule HC-L before 2015 and the revised Schedule HC-R from 2015 onward. The pre-2015 calculation treats credit-card commitments as unconditionally cancelable and applies a larger conversion factor to the remaining short-term commitments.

Component	Reporting items and treatment
Tier 1 capital	BHCK8274 through 2013:Q4; BHCA8274 from 2014:Q1.
On-balance-sheet assets	BHCKA224 through 2013:Q4; BHCAA224 from 2014:Q1.
Reported SLR	BHCAH036; positive reported values provide the calibration benchmark.
Derivatives	BHCEA167 through 2014:Q4; BHCKS542 + BHCKS549 from 2015:Q1. The earlier series is rescaled at the reporting change.
Other off-balance-sheet exposures, through 2014:Q4	Sum of BHCEB546, BHCE6570, BHCE3411, BHCE3429, BHCEA250, BHCEB541, BHCEB675, BHCEB681, BHCE6572, and BHCEG591.
Other off-balance-sheet exposures, from 2015:Q1	Sum of BHCKD992, BHCKD998, BHCKG607, BHCKG613, BHCKG619, BHCKS526, and BHCKG625. Morgan Stanley's BHCKG619 is adjusted as described above.
Repo-style gross balances	Sum of BHDMB987, BHCKB989, BHDMB993, and BHCKB995.
Repo-style counterparty exposure	BHCKS516 from 2015:Q1. Earlier exposure is imputed using the 2015:Q1–2016:Q2 mean exposure-to-gross-balances ratio; BHCE3433 is the fallback.

Table B.3: FR Y-9C inputs for Tier 1 capital and leverage exposure. [Back to text]

Notes: The supplementary leverage ratio (SLR) divides Tier 1 capital by total leverage exposure. The alphanumeric identifiers name the fields in the FR Y-9C bank regulatory filings used to measure each component. *Sources:* FR Y-9C bank regulatory filings.

Component	Reporting items and treatment
Total unused commitments, U	Sum of BHCK3814, BHCKJ455, BHCKJ456, BHCK3816, BHCK6550, BHCK3817, BHCKJ457, BHCKJ458, and BHCKJ459.
Short-term and cancelable commitments, S	$\max\{U - \text{BHCK6572} - \text{BHCKG591}, 0\}$. The subtracted items cover commitments over one year and commitments to ABCP conduits.
Cancelable portion, C	$\min\{\text{BHCKJ455} + \text{BHCKJ456}, S\}$.
Addition through 2014:Q4	$0.1C + 0.2(S - C)$.
Addition from 2015:Q1	$0.1 \times$ BHCKS540. Other reported credit-equivalent commitments enter the off-balance-sheet sum in Table B.3.

Table B.4: Commitment adjustment in the SLR exposure proxy. [Back to text]

Notes: Reported commitments from the FR Y-9C are converted to leverage-exposure amounts using the indicated conversion factors. The adjustment enters the denominator of the supplementary leverage ratio proxy. The symbols U, S, and C denote total unused, short-term and cancelable, and cancelable commitments, respectively. ABCP denotes asset-backed commercial paper. *Sources:* FR Y-9C bank regulatory filings.

Capital basis in earnings-call transcripts. I use disclosures from earnings-call transcripts identified as applying the final SLR rule. For each bank with overlapping disclosures,

I estimate the fully phased-in capital adjustment by dividing the difference between the reported FR Y-9C SLR and the earnings-call SLR by the fraction of deductions not yet phased in, then averaging across overlap quarters. The fractions of capital deductions not yet phased in are 0.8 in 2014, 0.6 in 2015, 0.4 in 2016, and 0.2 in 2017. I add the estimated adjustment times the remaining fraction to the earnings-call SLR. Disclosures already on a transitional basis enter unchanged; comparisons requiring an unavailable adjustment are omitted.

Goldman Sachs and Morgan Stanley. Before 2016:Q3, I multiply these banks' calibrated FR Y-9C proxies by the ratio of the adjusted earnings-call SLR to the proxy in quarters with usable earnings-call disclosures. I interpolate the factor linearly between quarters with earnings-call disclosures. Before the first quarter with an earnings-call disclosure, I use the mean factor from the first two quarters with earnings-call disclosures; after the final quarter with an earnings-call disclosure and before 2016:Q3, I retain the last factor. From 2016:Q3 onward, the binding-rule calculation uses the reported SLR.

Fit to reported SLRs. The calibrated proxy has a mean absolute error of 0.06 percentage points over 2016:Q3–2019:Q4. Calibrating only on 2016:Q3–2017:Q4 gives a mean absolute error of 0.07 percentage points in the held-out 2018:Q1–2019:Q4 observations.

C Calibration and additional estimation results

C.1 Equity funding cost calibration

Back

Table C.1 summarizes the literature used to calibrate the equity funding premium c_E. The assumed funding premium of $c_E = 0.05$ implies 5 basis points of additional funding cost per percentage point of required equity. Published loan-rate estimates range from 2.5 to 13 basis points and differ in assumptions about Modigliani–Miller offsets, taxes, and pass-through to loan rates. Loan-rate effects equal funding-cost effects under full pass-through; the estimates therefore provide calibration evidence rather than estimates of an identical parameter across studies.

Source	Loan-rate increase (basis points per pp of capital)	Assumptions
Kashyap et al. (2010)	2.5–4.5	Partial Modigliani–Miller offset; tax wedge
This paper ($c_E = 0.05$)	5	Funding cost before pass-through to loan rates
Miles et al. (2013)	≈ 5.5	45 percent Modigliani–Miller offset
Firestone et al. (2017)	3.9–7.8	50–100 percent pass-through to loan rates
Brooke et al. (2015)	5–10	≈ 53 percent Modigliani–Miller offset
BCBS (2010)	13	No Modigliani–Miller offset

Table C.1: Published estimates used to calibrate the equity funding premium. [Back to text]

Notes: The assumed equity funding premium $c_E = 0.05$ is the additional annual cost of equity relative to debt and implies 5 basis points of additional funding cost per percentage point of required equity. Loan-rate effects equal funding-cost effects under full pass-through; estimates differ in assumptions about taxes, Modigliani–Miller offsets, and pass-through.

C.2 Tests of equal supply elasticities

Back

Null hypothesis	Baseline p-value
All three elasticities are equal	0.068
Securities = mortgages	0.118
Securities = RW1 loans	0.031
Mortgages = RW1 loans	0.206

Table C.2: Tests of equality of asset supply elasticities. [Back to text]

Notes: Tests use the baseline substitution-corrected specification, which pools RW0 and RW0.2 securities and estimates separate elasticities for RW0.5 mortgages and RW1 loans. RW denotes risk weight. Estimates pool quarters 16–20 and include asset–date–horizon fixed effects. The joint test imposes equality of all three elasticities; each pairwise test imposes equality of the two named elasticities. Tests use bank-clustered inference with 28 banks. Pairwise p-values are unadjusted for multiple testing. *Sources:* FR Y-9C bank regulatory filings.

C.3 Pre-trends in portfolio shares

Back

Figure C.1 extends the portfolio-share response estimates to quarters before the change in required equity. The pre-event coefficients test for portfolio-share changes preceding the regulatory event; a lack of significant pre-event coefficients does not establish the identifying assumption.

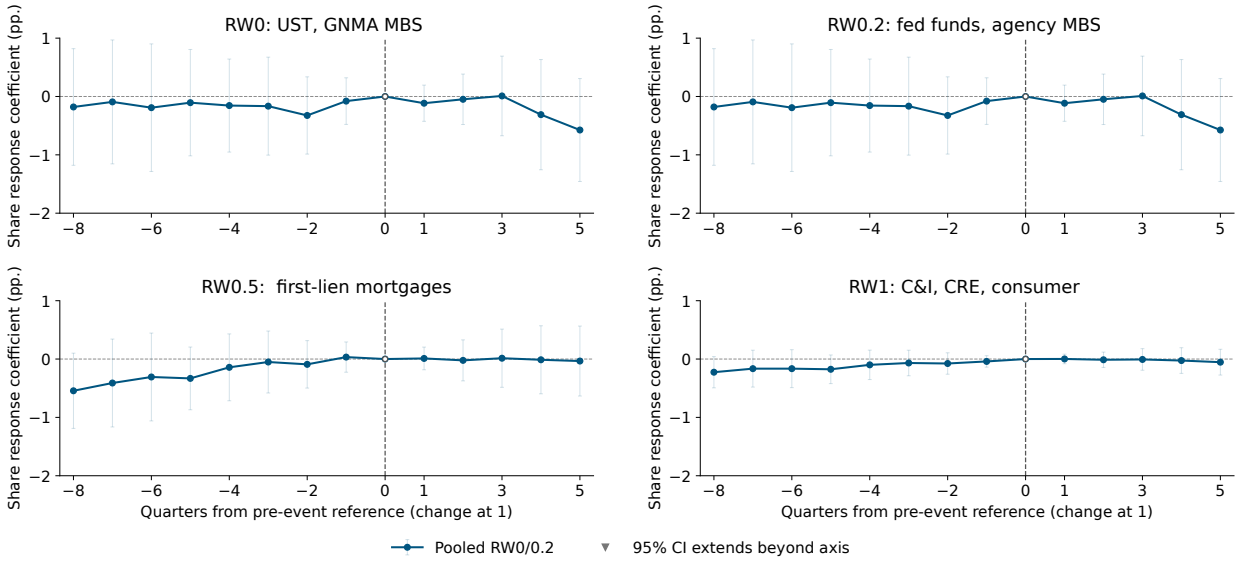


Figure C.1: Pre-period portfolio-share responses under discrete binding rules. [Back to text]

Notes: Points show portfolio-share responses (percentage points) to a one-percentage-point increase in required equity, by risk-weight category and horizon. The substitution-corrected specification pools elasticities for risk weights 0 and 0.2. The regulatory event occurs in quarter t. The horizontal axis counts quarters from the reference quarter $t-1$, shown at 0 with a zero coefficient. The required-equity change occurs at 1; negative values precede the reference quarter. Error bars show 95 percent confidence intervals clustered by bank; triangles mark intervals extending beyond the axis. *Sources:* FR Y-9C bank regulatory filings.

C.4 Controls for lagged portfolio shares

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Table C.3 compares controls for the starting portfolio share, $s_{ab,t-1}$, with controls for four lagged shares, $s_{ab,t-1}, \ldots, s_{ab,t-4}$. After the full-sample baseline, the remaining columns use identical observations. Comparing these columns distinguishes conditioning on the bank's initial allocation from conditioning on its recent portfolio trajectory. The final column also controls for past required-equity changes in four-quarter bins.

	Baseline	Baseline common sample	Ctrl. for s_{t-1}	Ctrl. for $s_{t-1}, \ldots, s_{t-4}$	Ctrl. for $s_{t-1}, \ldots, s_{t-4}$ + 4q policy bins from $t-8$ to $t-1$
RW0 / 0.2	30.49***	27.17**	21.56*	22.22*	24.17**
	(10.99)	(11.13)	(10.91)	(10.90)	(10.72)
RW0.5	11.62*	10.66*	8.56	10.45*	11.22**
	(5.83)	(6.01)	(5.56)	(5.62)	(5.30)
RW1	4.41*	5.94**	5.24**	5.03*	5.88**
	(2.48)	(2.61)	(2.54)	(2.73)	(2.70)
Observations	17596	13108	13108	13108	13108
Banks	28	27	27	27	27
Adjusted R^2	0.13	0.17	0.24	0.26	0.26

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.3: Substitution-corrected supply elasticities with lagged portfolio-share controls. [Back to text]

Notes: RW denotes risk weight. Estimates pool RW0 and RW0.2, measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects; the additional effects here change baseline fit statistics but not baseline elasticity estimates. Columns 2–5 use a common sample. In the headings, $s_{t-1}, \ldots, s_{t-4}$ denote the asset's portfolio shares in the four quarters before regulatory-event quarter t; 4q denotes four-quarter intervals, and Ctrl. denotes controls. The last column adds required-equity changes over quarters 1–4 and 5–8 before the event. All control coefficients vary by asset. *Sources:* FR Y-9C bank regulatory filings.

C.5 Required equity evaluated on lagged portfolios

Back

Table C.4 evaluates old and new regulation on the same prior-quarter portfolio. Let $K_a(P, X)$ denote asset a's required equity when regulation is P and the binding rule is calculated using portfolio and regulatory exposures X. The constructed change is

$$Z_{abt} = K_a(P_t, X_{b,t-1}) - K_a(P_{t-1}, X_{b,t-1}).$$

Here Z_{abt} is the change in required equity for asset a at bank b in event quarter t, holding the preceding-quarter portfolio and regulatory exposures fixed. The calculation recomputes the binding rule under each regulation. A portfolio change alone generates no shock; a regulatory change can alter required equity with or without changing the binding rule. For each event quarter t, both terms in Z_{abt} use the portfolio and regulatory exposures observed at $t-1$. Changes in the portfolio between $t-2$ and $t-1$ do not enter Z_{abt} as a separate source of required-equity changes. Constructed shocks are not smoothed; actual required

equity retains the baseline smoothing. Policy dates, observed portfolio-share outcomes, and exclusions of SCB changes as starting events remain unchanged.

The final two columns of Table C.4 add past and then past-and-future policy controls to the lagged-portfolio specification. The controls for past required-equity changes use each earlier event's own prior-quarter portfolio. Future increments instead evaluate old and new regulation on the starting event's fixed portfolio $X_{b,t-1}$, so future portfolio responses do not enter these controls. Past required-equity controls sum changes over quarters 1–4 and 5–8 before the event. Future controls sum regulatory changes over successive four-quarter intervals after the event, ending at the response horizon. Each interval enters through a common component, a component proportional to the asset's risk weight, and an RW0-specific component, with coefficients varying by asset. All columns use their largest valid samples; differences across columns can reflect both specification and sample changes. These regressions replace actual required-equity changes with constructed regulatory changes; they are not instrumental-variable estimates.

	Baseline (actual required equity)	Required equity evaluated on lagged portfolio	Lagged portfolio Ctrl. for 4q bins $t-8$ to $t-1$	Lagged portfolio Ctrl. for 4q bins $t-8$ to $t+h$ Fixed future portfolio
RW0 / 0.2	30.49***	25.41**	20.84	18.42
	(10.99)	(11.07)	(12.46)	(15.19)
RW0.5	11.62*	8.66*	10.16	28.36
	(5.83)	(4.68)	(5.96)	(49.76)
RW1	4.41*	3.30	5.37*	8.61
	(2.48)	(2.34)	(2.78)	(5.60)
Observations	17596	17236	12748	12728
Banks	28	28	26	25
Adjusted R^2	0.13	0.14	0.19	0.22

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.4: Supply elasticities with actual and lagged-portfolio required equity, with and without policy controls. [Back to text]

Notes: In the headings, t is the regulatory-event quarter, h is the response horizon in quarters, 4q denotes four-quarter intervals, and Ctrl. denotes controls. RW denotes risk weight. Estimates pool RW0 and RW0.2, measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects; the additional effects here change baseline fit statistics but not baseline elasticity estimates. Columns 2–4 evaluate the event-quarter regulatory change on the preceding quarter's balance sheet, excluding switches caused solely by portfolio changes. Past controls use each earlier event's preceding-quarter balance sheet. Future controls use the starting event's preceding-quarter balance sheet. Required-equity changes are summed within four-quarter intervals. Each column uses its largest valid sample.
Sources: FR Y-9C bank regulatory filings.

C.6 Excluding pandemic SLR exemption start/end events

Table C.5 compares estimates including and excluding the introduction and expiration of the temporary SLR exemption, in 2020:Q2 and 2021:Q2. The exclusion removes these quarters as requirement-change dates; it retains other dates and their subsequent outcomes. The table reports the three-group specification. The substitution-corrected elasticities remain positive in every asset category and are larger when the exemption events are excluded.

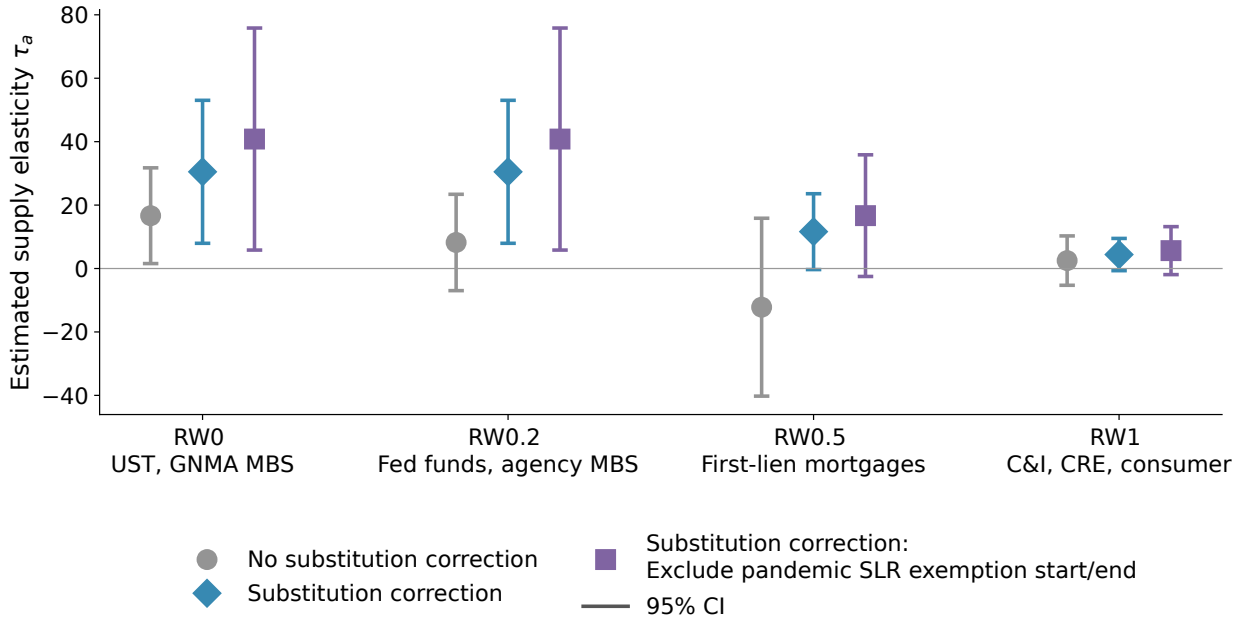


Figure C.2: Supply elasticities including and excluding pandemic SLR exemption events. [Back to text]

Notes: Gray circles and blue diamonds reproduce the main estimates without and with substitution correction. Purple squares apply the substitution correction after excluding 2020:Q2 and 2021:Q2 as requirement-change dates. Portfolio-share observations in those quarters remain available as outcomes for other starting dates. Corrected estimates pool RW0 and RW0.2; all estimates pool quarters 16–20 and use an equity funding premium of $c_E = 0.05$. Error bars are 95 percent bank-clustered confidence intervals. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

Asset	Include exemption events	Exclude exemption events
RW0	30.5***	40.8**
	(11.0)	(17.1)
RW0.2	30.5***	40.8**
	(11.0)	(17.1)
RW0.5	11.6*	16.7*
	(5.8)	(9.3)
RW1	4.4*	5.6
	(2.5)	(3.7)

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.5: Supply elasticities with and without pandemic SLR exemption events: discrete binding rules. [Back to text]

Notes: Estimates pool quarters 16–20. Include retains 2020:Q2 and 2021:Q2 as requirement-change dates; Exclude removes those two dates. Portfolio-share observations in those quarters remain available as outcomes for other starting dates. All specifications exclude reserves and vault cash. Both columns include asset–date–horizon fixed effects; the exclusion column also includes bank–date effects. The three-group specification pools RW0 and RW0.2 and estimates separate elasticities for RW0.5 and RW1. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

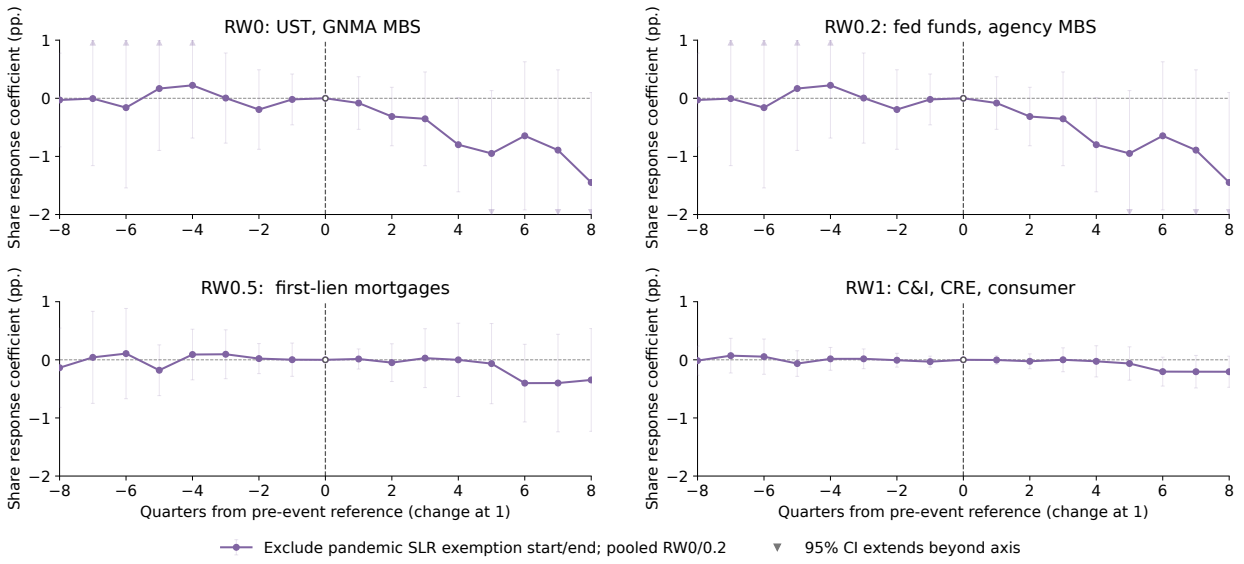


Figure C.3: Pre-trends excluding pandemic SLR exemption start/end events.

Notes: The specification excludes 2020:Q2 and 2021:Q2 as requirement-change dates, as in the exclusion column of Table C.5. Portfolio-share observations in those quarters remain available as outcomes for other starting dates. Substitution weights are proportional to the supply elasticities estimated after excluding those requirement-change dates. Quarter zero is the pre-event reference quarter; the required-equity change occurs at displayed quarter one. Error bars show 95 percent bank-clustered confidence intervals conditional on the substitution weights; triangles mark intervals extending beyond the axis. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

C.7 Controls for policy leads and lags

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Table C.6 adds controls for past required-equity changes in two bins, covering quarters 1–4 and 5–8 before the event. The final column also controls for future regulatory changes in successive four-quarter intervals through the response horizon, evaluating each change on the starting event's preceding-quarter portfolio. The event-quarter change and past controls continue to use actual required equity. Each four-quarter interval enters through a common component, a component proportional to the asset's risk weight, and an RW0-specific component, with coefficients varying by asset. The reported supply elasticities describe the response to the event-quarter change conditional on these policy controls.

	Baseline	Ctrl. for 4q bins from $t-8$ to $t-1$	Ctrl. for 4q bins from $t-8$ to $t+h$ Fixed future portfolio
RW0 / 0.2	30.49***	26.78**	24.07
	(10.99)	(11.44)	(14.79)
RW0.5	11.62*	12.21**	19.00
	(5.83)	(5.87)	(12.43)
RW1	4.41*	6.80**	8.74**
	(2.48)	(2.71)	(3.99)
Observations	17596	13168	12928
Banks	28	27	26
Adjusted R^2	0.13	0.18	0.22

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.6: Substitution-corrected supply elasticities controlling for leads and lags of required-equity changes. [Back to text]

Notes: In the headings, t is the regulatory-event quarter, h is the response horizon in quarters, 4q denotes four-quarter intervals, and Ctrl. denotes controls. RW denotes risk weight. Estimates pool RW0 and RW0.2, measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects; the additional effects here change baseline fit statistics but not baseline elasticity estimates. Event-quarter and past required-equity changes use contemporaneous portfolios. Future regulatory changes use the starting event's preceding-quarter balance sheet. Controls sum required-equity changes within four-quarter intervals. Each column uses its largest valid sample. *Sources:* FR Y-9C bank regulatory filings.

C.8 Alternative inference

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	Baseline	Bank WCB	Bank + quarter clustering	Bank + quarter WCB
RW0 / 0.2	30.49***	30.49*	30.49***	30.49*
SE	(10.99)	(10.99)	(8.48)	(8.48)
RW0.5	11.62*	11.62*	11.62**	11.62*
SE	(5.83)	(5.83)	(4.88)	(4.88)
RW1	4.41*	4.41	4.41***	4.41*
SE	(2.48)	(2.48)	(1.59)	(1.59)
Observations	17596	17596	17596	17596
Banks	28	28	28	28
Adjusted R^2	0.13	0.13	0.13	0.13

Standard errors clustered by bank or by bank and event quarter, as indicated, in parentheses. Bootstrap columns use bootstrap p-values.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.7: Alternative inference for the baseline three-group specification. [Back to text]

Notes: RW denotes risk weight. Estimates pool RW0 and RW0.2, measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects; the additional effects here change baseline fit statistics but not baseline elasticity estimates. All columns report the baseline estimates. WCB denotes restricted wild cluster bootstrap-t inference with 999 bank-level Rademacher draws and nonlinear refitting. Two-way covariance uses inclusion–exclusion, with negative eigenvalues set to zero; two-way WCB uses bank-and-starting-quarter studentization. *Sources:* FR Y-9C bank regulatory filings.

C.9 Alternative asset-pooling specifications

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The substitution-corrected estimates in Table C.8 use three alternative restrictions on the asset-specific response coefficients β_a. Since $\tau_a = -\beta_a/c_E$, each restriction also applies to the supply elasticities.

Pooling securities only. The specification used in the main results assigns a common coefficient to RW0 and RW0.2 assets and estimates separate coefficients for RW0.5 and RW1 assets:

$$\beta_{\text{RW0}} = \beta_{\text{RW0.2}} = \beta_L, \qquad \beta_{\text{RW0.5}} = \beta_M, \qquad \beta_{\text{RW1}} = \beta_H.$$

The coefficients β_L, β_M, and β_H correspond to securities, first-lien mortgages, and RW1 assets, respectively.

Pooling securities and pooling loans. The two-coefficient specification retains the common coefficient for RW0 and RW0.2 assets and also assigns a common coefficient to

RW0.5 and RW1 assets:

$$\beta_{\text{RW0}} = \beta_{\text{RW0.2}} = \beta_L, \qquad \beta_{\text{RW0.5}} = \beta_{\text{RW1}} = \beta_H.$$

Linear in risk weight. The linear specification estimates an intercept and a slope relating the response coefficient to the asset's risk weight w_a:

$$\beta_a = \beta_0 + \beta_1 w_a.$$

Here β_0 is the intercept, β_1 is the slope, and w_a is asset a's standardized risk weight. I report the converged solution with the lowest residual sum of squares among solutions with a positive sum of supply elasticities. Table C.8 compares supply elasticities under these three restrictions with estimates that omit the portfolio-substitution correction and allow a separate coefficient for each risk-weight category. Response coefficients satisfy $\widehat{\beta}_a = -c_E \widehat{\tau}_a$; their standard errors equal c_E times the reported elasticity standard errors.

	No substitution correction	Pooled RW0/0.2	Pooled RW0/0.2 and RW0.5/1	Linear $\beta_0 + \beta_1 RW$
RW0	16.65**	30.49***	31.36***	25.49***
	(7.36)	(10.99)	(11.25)	(8.38)
RW0.2	8.21	30.49***	31.36***	21.29***
	(7.40)	(10.99)	(11.25)	(6.76)
RW0.5	−12.19	11.62*	5.54**	14.99***
	(13.67)	(5.83)	(2.39)	(4.47)
RW1	2.50	4.41*	5.54**	4.49*
	(3.79)	(2.48)	(2.39)	(2.63)
N. Banks	28	28	28	28
N. Quarters	42	42	42	42
SE clustering	Bank	Bank	Bank	Bank
Adjusted R^2	0.13	0.11	0.11	0.11
Adjusted within R^2	0.00	0.00	0.00	0.00
Bank–date FE	Yes	No	No	No
Asset–date–horizon FE	Yes	Yes	Yes	Yes

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.8: Supply elasticities under alternative asset-pooling specifications. [Back to text]

Notes: RW denotes risk weight. Pooled RW0/0.2 imposes a common securities elasticity; two groups additionally pools RW0.5 and RW1. Linear in RW restricts elasticity to an intercept and slope in risk weight. No correction omits portfolio substitution. Estimates measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event. Elasticities equal $-\widehat{\beta}_a/c_E$, where $\widehat{\beta}_a$ is the portfolio-share response coefficient and $c_E = 0.05$ is the equity funding premium. *Sources:* FR Y-9C bank regulatory filings.

C.10 Substitution into assets outside the four categories

Back

The four asset categories do not exhaust the banking-book denominator. The baseline substitution correction treats the remaining assets as having a zero supply response. Table C.9 checks whether the estimated elasticities depend on this restriction by adding those remaining assets as a fifth category, the outside good, and estimating its supply elasticity. All columns retain the baseline dollar amounts of assets in the four risk-weight categories and the same banking-book denominator, including its exclusion of reserves and vault cash. The four categories and the outside good therefore exhaust that denominator.

The first column of Table C.9 reports the baseline elasticities with a zero supply elasticity for the outside good. The two outside-good specifications measure a proxy for the outside good's average risk weight from FR Y-9C Schedule HC-R and allow its portfolio share to respond to regulatory costs. Both outside-good specifications calculate the outside good's

average risk weight as its risk-weighted assets divided by the dollar amount of assets in the outside good. They measure that average risk weight before the regulatory-change quarter and hold it fixed when calculating the change in required equity under the bank's binding rule. The substitution correction then includes the estimated outside elasticity. Treating the heterogeneous outside assets as one category remains an aggregation assumption: their average risk weight does not necessarily capture differences in supply responses across those assets.

Outside risk weight from asset components. I divide assets outside the four risk-weight categories into loans, securities, and cash balances. For omitted loans, I use the ratio of reported risk-weighted assets to carrying values in HC-R's two ordinary "all other loans" rows as a proxy for their average risk weight. For omitted securities, I subtract the estimated risk-weighted assets of the included securities from reported securities risk-weighted assets, including separately reported securitizations when available. The remaining cash balances use HC-R cash risk-weight allocations. Before 2015, I apply each bank's median component risk weights over 2015:Q4–2017:Q4 to its earlier component holdings, requiring at least four valid calibration quarters. This permits earlier composition changes but assumes that the component risk weights can be carried backward across the reporting change. From 2015 onward, I use the available lagged measurements directly.

Outside risk weight from reported RWA residual. I instead measure outside-loan risk-weighted assets by subtracting the estimated risk-weighted assets of the included loans from total reconstructed loan risk-weighted assets. Securities and cash follow the same approach as in the asset-component specification. This specification uses the available HC-R measurements in each reporting regime rather than carrying later component weights backward. It relies more directly on the accuracy of the included-loan mapping and on the coverage of earlier HC-R reports. Observations with negative loan or securities residuals are excluded. Both checks retain approximations in the securities mapping and, where necessary, the allocation of jointly reported securitization risk-weighted assets.

The estimates for the four asset categories remain close to the baseline re-estimated on the same observations in Panel B. The outside elasticities are positive but imprecisely estimated. The samples differ because the checks require usable outside-asset measurements. Relative to the baseline, the asset-component specification excludes SVB Financial Group, KeyCorp, and Huntington; the reported-RWA-residual specification excludes SVB Financial Group and KeyCorp. The securities risk-weighted-asset residuals exclude SVB Financial Group and KeyCorp. Huntington lacks enough valid 2015:Q4–2017:Q4 observations to impute its earlier component risk weights and has no usable later observations in the asset-component specification.

	Baseline: fixed outside good	Outside RW from asset components	Outside RW from reported RWA residual
A. Supply elasticities			
RW0 and RW0.2	30.49***	28.06**	32.28***
	(10.99)	(11.11)	(10.93)
RW0.5	11.62*	12.37*	11.91*
	(5.83)	(6.60)	(5.84)
RW1	4.41*	5.50**	4.43**
	(2.48)	(2.27)	(2.09)
Outside good	0 (fixed)	0.47	5.12
		(7.03)	(11.19)
Bank–quarter–horizon observations	4,399	3,656	3,468
Banks	28	25	26
B. Baseline on each column's sample			
RW0 and RW0.2	30.49***	27.99**	32.63***
	(10.99)	(11.05)	(10.56)
RW0.5	11.62*	12.51*	12.38*
	(5.83)	(6.85)	(6.05)
RW1	4.41*	5.54**	4.68*
	(2.48)	(2.59)	(2.31)

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.9: Supply elasticities allowing substitution into the outside good. [Back to text]

Notes: RW denotes risk weight; RWA denotes risk-weighted assets. The outside good comprises banking-book assets outside the four modeled risk-weight categories. All columns use the banking-book denominator, pool horizons 16–20, impose a common elasticity for RW0 and RW0.2, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects. Panel A fixes the outside elasticity at zero in the baseline and estimates it in the two outside-good specifications. Panel B re-estimates the original baseline on each column's sample. Observations count bank–quarter–horizon combinations, before stacking asset categories. Inference treats the measured risk-weight proxies and equity funding premium as fixed. *Sources:* FR Y-9C bank regulatory filings.

C.11 Risk-weight allocations reported in Schedule HC-R

Back

Table C.10 compares the baseline mapping with banks' reported risk-weight allocations in Schedule HC-R, Part II. I apply the reported fractions within ordinary loan and securities categories to the same modeled dollar amounts used in the baseline. Both specifications use identical observations, with regulatory-change quarters from 2015 onward and responses pooled over quarters 16–20. Reverse-repo allocations, the treatment of cash, the portfolio-share denominator, and asset-specific required equity retain their baseline construction. The comparison isolates differences in the risk-weight mapping within the modeled loan and securities categories.

	Category mapping	HC-R fractions
RW0 / 0.2	33.23***	37.23***
	(11.06)	(11.07)
RW0.5	8.36**	6.87**
	(3.29)	(2.53)
RW1	4.03*	4.85**
	(2.10)	(1.76)
Observations	10632	10632
Banks	28	28
Adjusted R^2	0.19	0.19

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.10: Supply elasticities using Schedule HC-R risk-weight allocations on a matched sample from 2015 onward. [Back to text]

Notes: Both columns use the three-group substitution correction and measure portfolio-share changes from the quarter before the regulatory event to quarters 16–20 after the event. HC-R denotes the FR Y-9C regulatory-capital schedule; RW denotes risk weight. HC-R fractions allocate the baseline loan and securities amounts across risk weights. Ordinary HC-R securities exclude separately reported securitization exposures. Both columns pool RW0 and RW0.2, use the same post-2015 sample and quarters 16–20, and include bank–date and asset–date–horizon fixed effects. The corrected specification in Table 3 omits bank–date effects. *Sources:* FR Y-9C bank regulatory filings.

C.12 Sensitivity to the equity funding premium

Back

Figure C.4 reports implied supply elasticities over alternative values of c_E. A larger equity funding premium implies a smaller supply elasticity for a given estimated portfolio-share response, since $\widehat{\tau}_a = -\widehat{\beta}_a / c_E$.

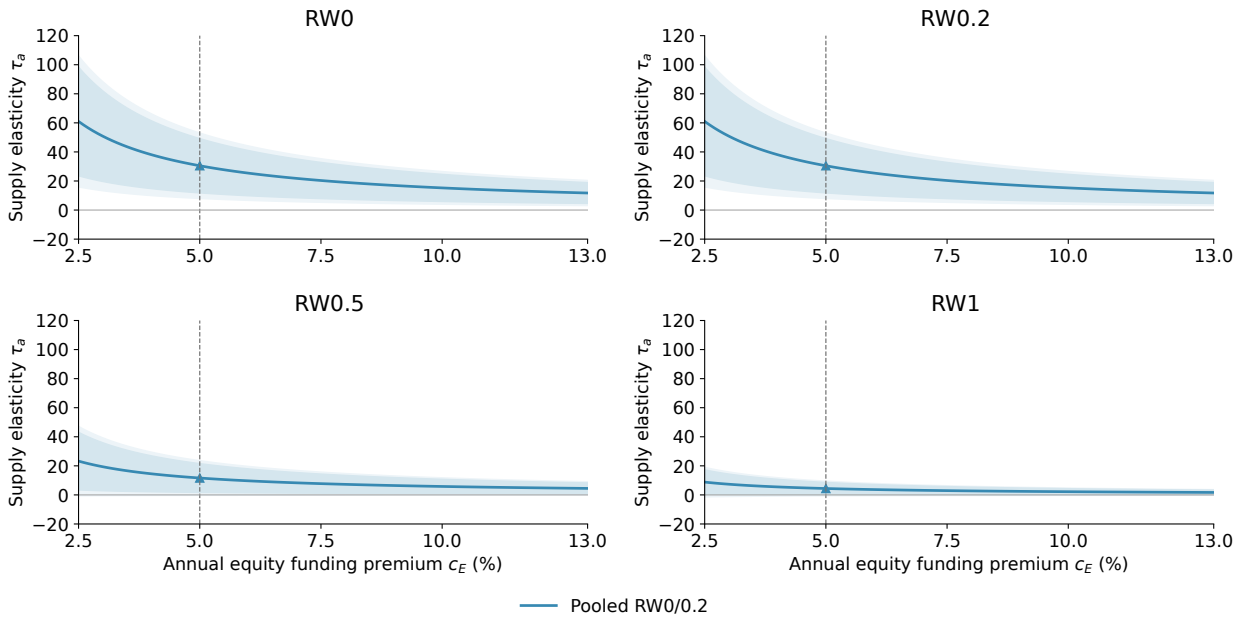


Figure C.4: Long-run supply elasticities under alternative equity funding premia. [Back to text]

Notes: Supply elasticities $\widehat{\tau}_a$ equal $-\widehat{\beta}_a/c_E$, where $\widehat{\beta}_a$ is the long-run portfolio-share response coefficient and c_E is the equity funding premium. The horizontal axis expresses the equity funding premium c_E as a percentage; the baseline value is $c_E = 0.05$. RW denotes risk weight. The three-group curves impose a common supply elasticity for RW0 and RW0.2 and separate elasticities for RW0.5 and RW1. Shaded bands show 90 percent and 95 percent confidence intervals clustered by bank, conditional on c_E. *Sources:* FR Y-9C bank regulatory filings.

C.13 Continuous weights on RBC and SLR requirements

Back

Banks may put weight on both capital rules when they are uncertain which will bind. I replace the required-equity measure based on a single binding capital rule with a weighted average of the RBC and SLR requirements:

$$k_{abt}^{Bind} = p_{bt}\ell_{bt}^{RBC}w_a + (1 - p_{bt})\ell_{bt}^{SLR}, \qquad p_{bt} = \Phi(\Delta_{bt}/\sigma_\Delta).$$

Here k_{abt}^{Bind} is asset a's required equity at bank b in quarter t, w_a is the asset's risk weight, and ℓ_{bt}^{RBC} and ℓ_{bt}^{SLR} are the bank-level required ratios. The weight p_{bt} applies to the RBC requirement. The buffer difference Δ_{bt} is the SLR buffer minus the RBC buffer, with both buffers measured as excess equity per dollar of risk-weighted assets. The function Φ is the standard normal cumulative distribution function. I estimate the buffer-uncertainty parameter σ_Δ jointly with the supply elasticities. A larger buffer under SLR relative to RBC puts more weight on RBC; a larger σ_Δ makes the weights less sensitive to the difference between buffers.

The continuous-weight estimates preserve the main ranking: securities respond more to required equity than loans. The three-group elasticity is 47.1 for RW0 and RW0.2, 16.9 for RW0.5, and 4.9 for RW1, compared with 30.5, 11.6, and 4.4 under discrete binding rules. The continuous-weight estimates are less precise. Figures C.5 and C.6 compare response coefficients and supply elasticities with the discrete-rule baseline; Table C.11 reports the continuous-weight estimates and standard errors. Figure C.7 reports portfolio-share responses over time under continuous rule weights.

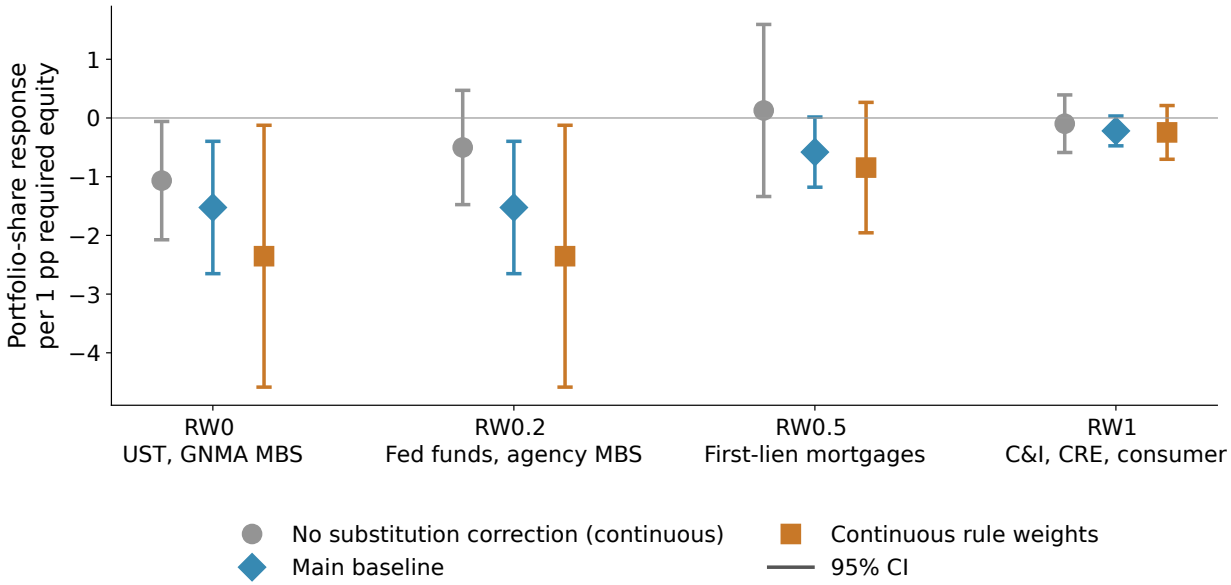


Figure C.5: Long-run portfolio-share responses with and without substitution correction under continuous rule weights. [Back to text]

Notes: Points show portfolio-share responses, in percentage points, to a one-percentage-point increase in required equity, pooling quarters 16–20. Blue diamonds show the main discrete-rule baseline. Orange squares show continuous-rule estimates with substitution correction and pooled RW0 and RW0.2; gray circles show continuous-rule estimates without substitution correction. Error bars give 95 percent bank-clustered confidence intervals. Both continuous regressions use the substitution-corrected estimate of the sensitivity of rule weights to RBC–SLR capital-buffer differences. Confidence intervals for the uncorrected regression treat this sensitivity as fixed. RBC denotes risk-based capital; SLR denotes supplementary leverage ratio; RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

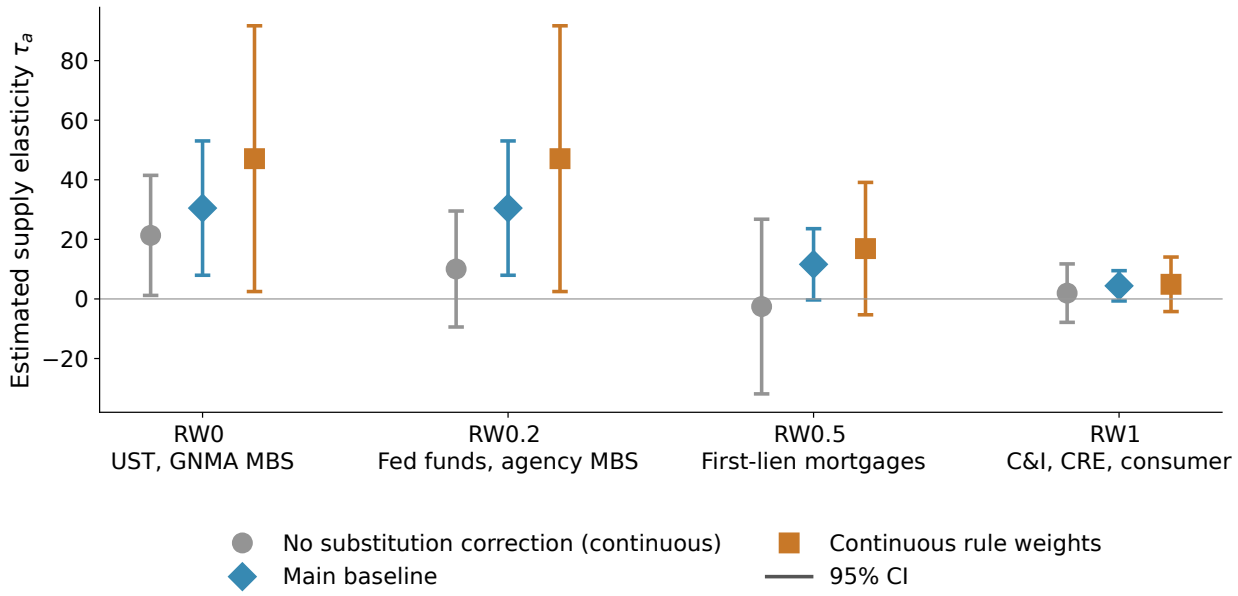


Figure C.6: Supply elasticities with and without substitution correction under continuous rule weights. [Back to text]

Notes: Points show supply elasticities $\widehat{\tau}_a$, pooling quarters 16–20 with an equity funding premium of $c_E = 0.05$. Blue diamonds show the main discrete-rule baseline. Orange squares show continuous-rule estimates with substitution correction and pooled RW0 and RW0.2; gray circles show continuous-rule estimates without substitution correction. Error bars give 95 percent bank-clustered confidence intervals. Both continuous regressions use the substitution-corrected estimate of the sensitivity of rule weights to RBC–SLR capital-buffer differences. Confidence intervals for the uncorrected regression treat this sensitivity as fixed. RBC denotes risk-based capital; SLR denotes supplementary leverage ratio; RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

	Response coefficient $\widehat{\beta}_a$	Supply elasticity $\widehat{\tau}_a$
RW0 / 0.2	-2.35^{**}	47.09^{**}
	(1.09)	(21.74)
RW0.5	-0.84	16.90
	(0.54)	(10.83)
RW1	-0.25	4.91
	(0.22)	(4.46)
Observations	17596	17596
Banks	28	28
Adjusted R^2	0.13	0.13

Standard errors clustered by bank in parentheses.
Two-sided tests: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table C.11: Response coefficients and supply elasticities with continuous rule weights. [Back to text]

Notes: Estimates pool quarters 16–20 and impose a common elasticity for RW0 and RW0.2. Required equity weights the risk-based capital and supplementary leverage ratio requirements using the difference between their capital buffers. Supply elasticities $\widehat{\tau}_a$ equal $-\widehat{\beta}_a/c_E$, where $\widehat{\beta}_a$ is the portfolio-share response coefficient and $c_E = 0.05$ is the equity funding premium. Inference conditions on c_E. RW denotes risk weight. *Sources:* FR Y-9C bank regulatory filings.

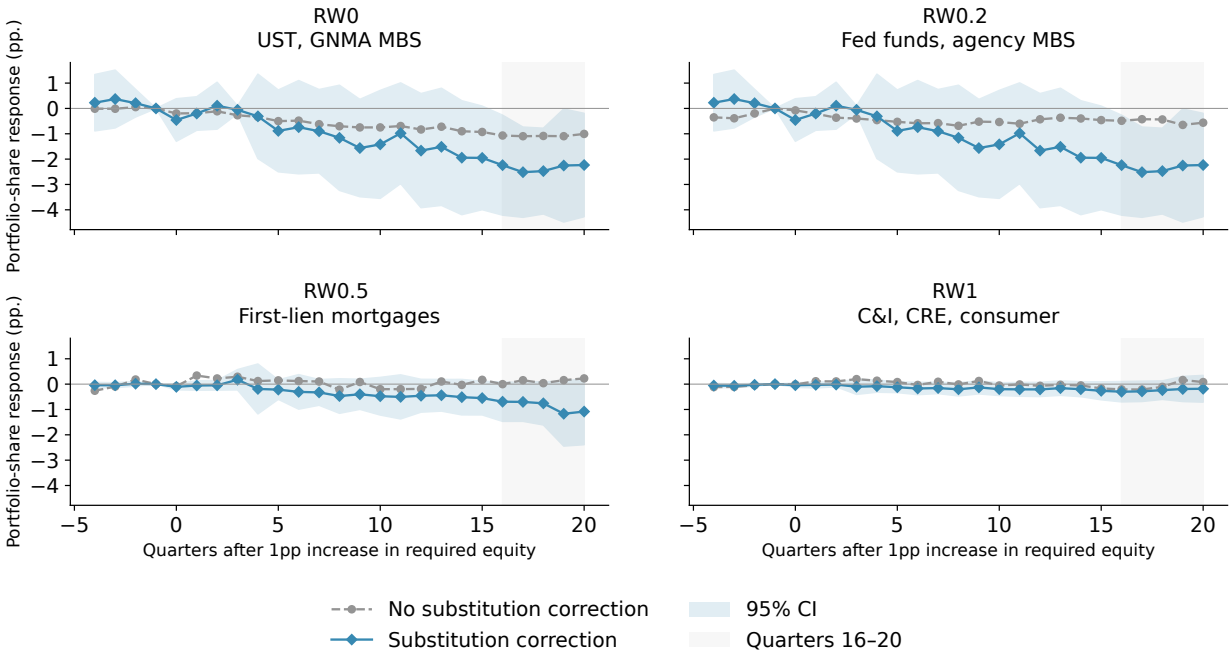


Figure C.7: Dynamic asset-share responses with continuous rule weights. [Back to text]

Notes: Lines show portfolio-share responses, in percentage points, to a one-percentage-point increase in required equity. Dashed gray lines with circles show estimates without substitution correction; solid blue lines with diamonds show estimates with portfolio-substitution correction that impose a common supply elasticity for RW0 and RW0.2 assets. RW denotes risk weight. Blue shading gives 95 percent confidence intervals clustered by bank. Quarter −1 is the zero reference point; quarter 0 includes the event-quarter response. Light-gray shading marks quarters 16–20, pooled in the long-run estimates. Required equity weights the risk-based capital (RBC) and supplementary leverage ratio (SLR) requirements using the difference between the bank's capital buffers under the two rules. Estimates use portfolio shares for 28 banks, excluding trading assets, reserves, and vault cash from the denominator, and control for bank–date and asset–date–horizon fixed effects. *Sources:* FR Y-9C bank regulatory filings.

D Uncertainty in the policy application

Table D.1 reports the observed and model-implied convergence slopes from Section 5. Each slope relates the change in a bank's average asset risk weight to its initial average risk weight, giving each bank equal weight. The percentage explained is 100 times the model-implied slope divided by the observed slope. Confidence intervals are reported for the underlying slopes; the percentage explained is reported as a point estimate.

For the aggregate portfolio changes and model-implied convergence slope, I apply the delta method to the full bank-clustered covariance matrix of the baseline supply elasticities. The 95 percent intervals use Student's t critical values with 27 degrees of freedom, holding historical portfolios, regulatory requirements, bank weights, the equity funding premium, and the application sample fixed. The observed-convergence interval uses heteroskedasticity-robust standard errors with the HC1 finite-sample correction and a Student's t critical value with 13 degrees of freedom.

	Observed convergence slope	Model-implied convergence slope	Percentage of convergence explained
Point estimate	-0.14	-0.05	34.6 percent
95 percent confidence interval	[-0.33, 0.05]	[-0.08, -0.01]	—

Table D.1: Observed and model-implied portfolio convergence. [Back to text]

Notes: The sample contains 15 banks with initial and final portfolio observations. Portfolio changes compare the 2011:Q1–2013:Q2 and 2022–2024 averages. Both regressions include an intercept and give each bank equal weight. Brackets report 95 percent confidence intervals. Observed-slope intervals use heteroskedasticity-robust standard errors; model-implied intervals reflect uncertainty in the bank-clustered supply-elasticity estimates. The percentage explained divides the model-implied slope by the observed slope. *Sources:* FR Y-9C bank regulatory filings.